Chapter 1: Concept of Computer networks

History of Internet
Concept of computer networks
Network architecture
Packet switching vs. circuit switching

Reading: Chapter 1, Computer Networks, Tanenbaum

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1

1

About me

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2

Course objectives

- Understand the Internet technology
  - Some networking mechanisms
  - Some protocols of TCP/IP
- Explain how the Internet works
- Be able to use the Internet efficiently, install new technologies and services

Topics

- Introduction to computer networks
- Basic concepts of computer communication model (OSI)
- Details of each layer
  - Physical Layer
  - Data-link Layer
  - Internet/ Network Layer
    - Routing problem
  - Transport Layer
  - Application Layer

Assessment

- Progress (50%)
  - Practical labs (60%)
  - Mid-term examination (40%)
  - Attendance (HUST regulation)
    - No absence: +1 point
    - Missing 3 or more: -1 point
  - Exercises: Students can earn extra points by solving exercises provided by the lecturer
- Final examination (50%)

5

History of the Internet

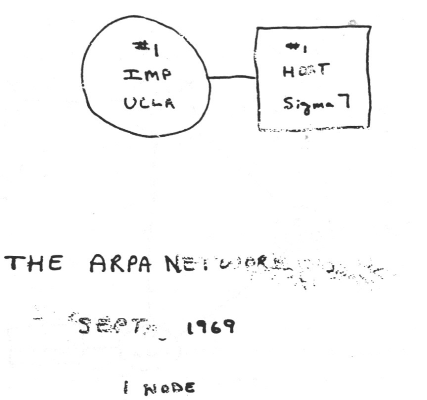


FIGURE 6.1 Drawing of September 1969
(Courtesy of Alex McKenzie)

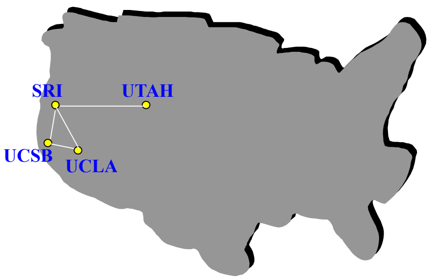
- Originated from an experiemental project of ARPA
- Intially having only two nodes (IMP atUCLA and IMP at SRI).

ARPA: Advanced Research Project Agency
UCLA: University California Los Angeles
SRI: Stanford Research Institute
IMP: Interface Message Processor, that \ each computer must be attached with

Source: http://www.cybergeography.org/atlas/historical.html

6

In 12/1969, after 3 months



A network with 4 nodes, 56kbps

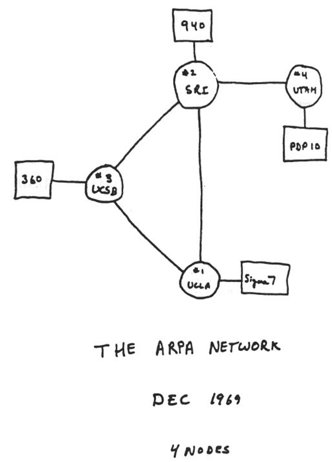
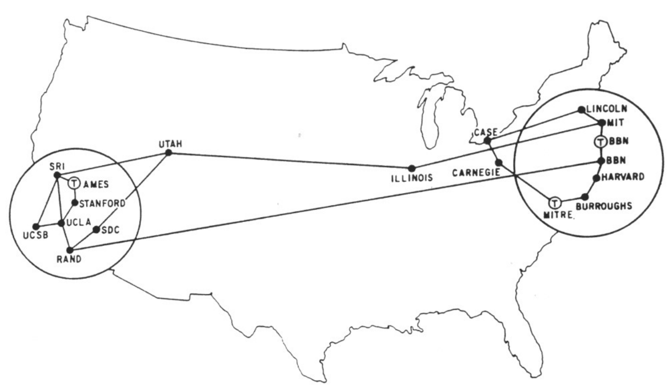


FIGURE 6.2 Drawing of 4 Node Network (Courtesy of Alex McKenzie)

UCSB:University of California, Santa Barbara
UTAH:University of Utah

source: http://www.cybergeography.org/atlas/historical.html

ARPANET, 1971



MAP 4 September 1971

Source: http://www.cybergeography.org/atlas/historical.html

One node was added each month

Years 70s: Interconnection, new network architectures and private architectures

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Expansion of ARPANET, 1974

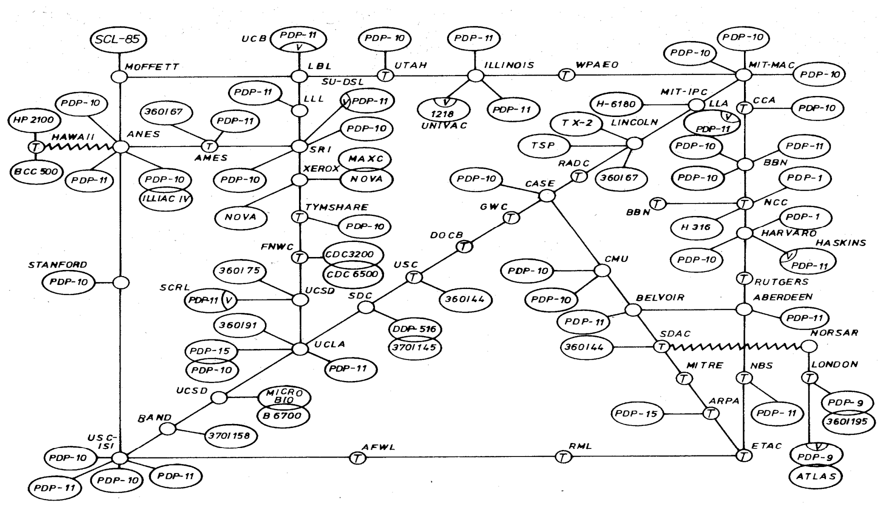

Abb. 4 ARPA NETwork, topologische Karte. Stand Juni 1974.

source: http://www.cybergeography.org/atlas/historical.html

Traffic each day not more than 3.000.000 package

Years 70s

- Since 1970, new networks private architectures appear:
  - ALOHAnet in Hawaii
  - DECnet, IBM SNA, XNA
- 1974: Cerf & Kahn – principles of interconnection of open systems (Turing Awards)
- 1976: Ethernet, Xerox PARC
- End of 1970s: ATM

Years 80s: New protocols, more expansion

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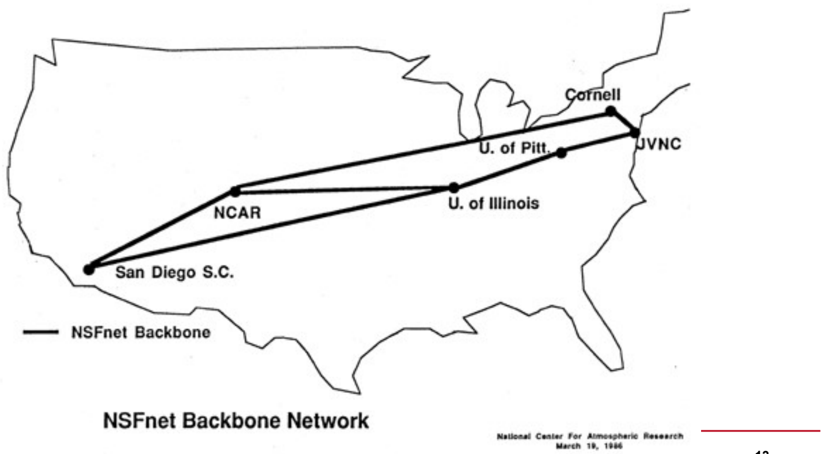
1981: Beginning of NSFNET

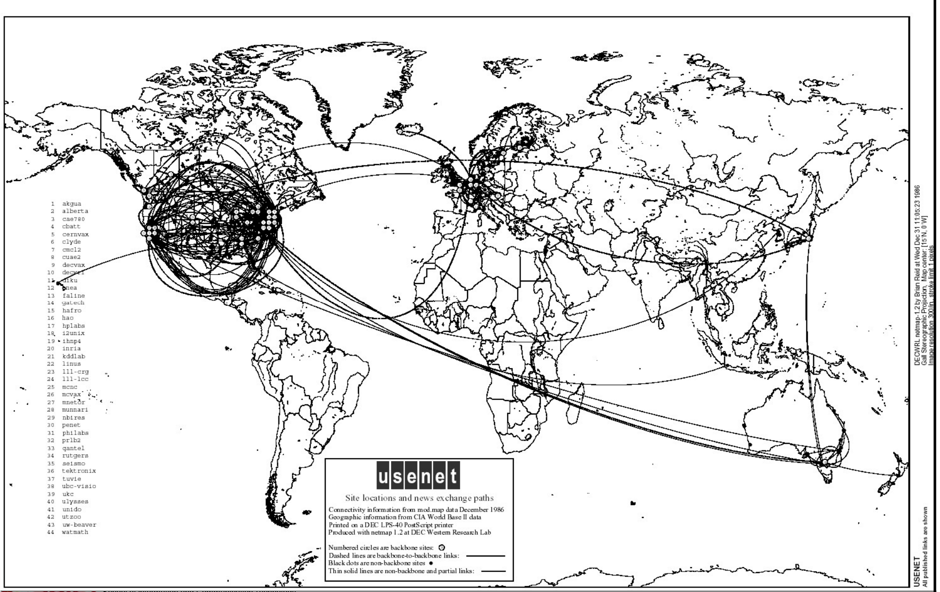
NSF: National Science Foundation

NSF network is separated from ARPANET for academic research



13

1986: Connect USENET and NSFNET



Source: http://www.cybergeography.org/atlas/historical.html

14

More network to join and more protocol

- More networks join in: MFENET, HEPNET (Dept. Energy), SPAN (NASA), BITnet, CSnet, NSFnet, Minitel ...
- TCP/IP is standardized and becomes popular in 1980
- Berkeley integrate TCP/IP in BSD Unix
- Services: FTP, Mail, DNS ...

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15

Years 90s: Web and E-commerce over Internet

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16

Years 90s

- Begining of 90s:
  Begining of Web
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, Netscape
- End of 90s:
  Commercialized the Internet

End of 1990's – 2000's:

- Many new Internet applications was introduced:
  - Chat, file sharing P2P...
  - E-commerce, Yahoo, Ebay, Amazon, Google...
- > 50 millions hosts, > 100 millions users.

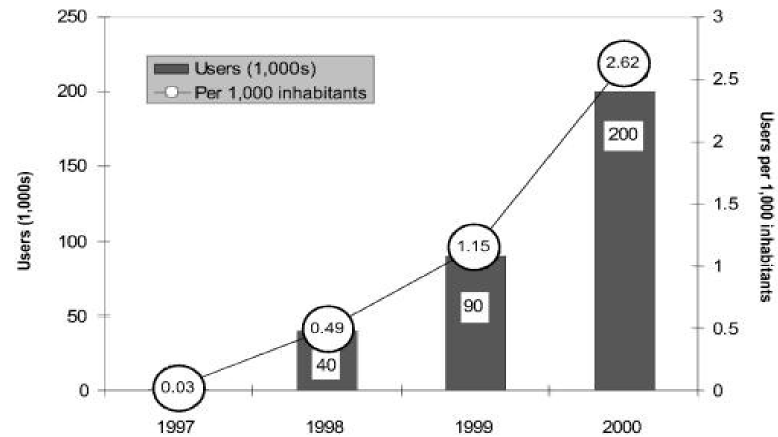
17

Internet in Việt Nam

- 1996: Preparation for the Internet infrastructure
  - ISP: VNPT
  - 64kbps, 01 connection to the world, few end users.
- 1997: Việt Nam connects to the Internet officially
  - 1 IXP (Internet Exchange Point): VNPT
  - 4 ISP (Internet Service Provider) : VNPT, Netnam (IOT), FPT, SPT
- 2007: After 10 years
  - 20 ISPs, 4 IXPs: VNPT, FPT, Viettel, EVN Telecom
  - 19 mil. users, 22.04% population

18

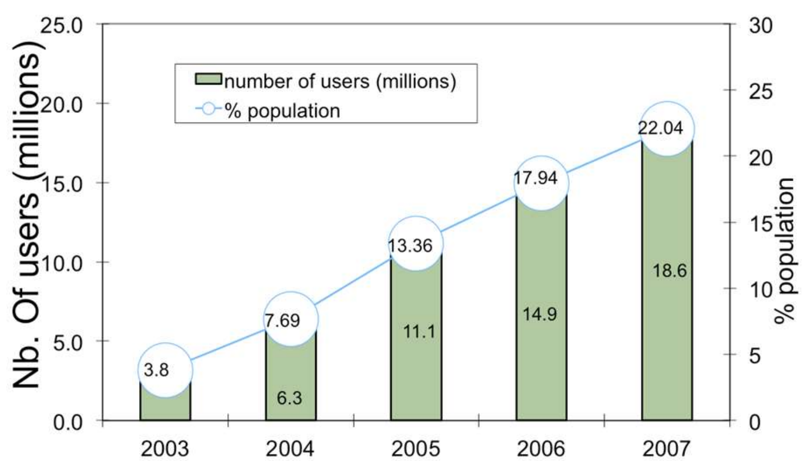
Development of the Internet in Vietnam



The numbers of users are estimated by 2 times the number of subscribers

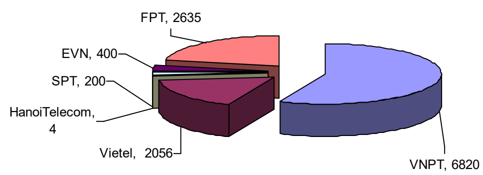
Source: Vietnam Internet Case Study, http://www.itu.int/asean2001/reports/material/VNM%20CS.pdf

Statistics until 2007



Source: *Vnnic*, http://www.thongkeinternet.vn

Bandwidth to the world (Mbps), 3rd Quarter



Total: 12115.0 Mbps

Fixed internet subscription, 2019

5 2019 Xem

Tình hình phát triển thuê bao băng rộng cố định tháng 5/2019	
Số thuê bao truy nhập Internet qua hình thức xDSL:	182,853
Số thuê bao truy nhập Internet qua kênh thuê riêng:	22,929
Số thuê bao truy nhập Internet qua hệ thống cáp truyền hình (CATV):	868,039
Số thuê bao truy nhập Internet qua hệ thống cáp quang tới nhà thuê bao (FTTH):	12,606,506
Tổng số thuê bao băng rộng cố định:	13,680,327

Statistics are provided by Department of Telecommunication, Ministry of Information and Communication.
http://vnta.gov.vn/thongke/Trang/dulieuthongke.aspx

Some fixed internet subscription technologies

- Dial-up:
  - 56kbps,
  - use public telephone lines,
  - Data are transmitted over the same frequency with voice,
  - Old technology, popular before 2000
- ADSL, xDSL:
  - few Mbps,
  - use public telephone lines,
  - Data are transmitted over the different frequency with voice,
  - popular between 2000-2010
- Internet over TV cable
  - Use TV cable to carry data
- FTTH
  - several dozen Mbps,
  - Use optical fiber
  - Popular nowadays.

23

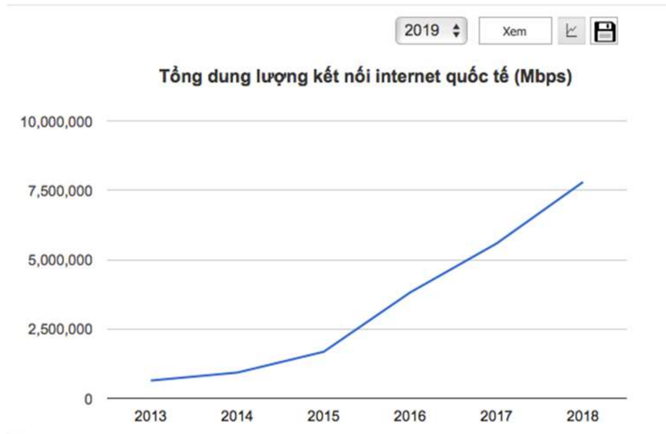
Data usage on mobile phones 2019

Tình hình phát triển thuê bao điện thoại di động tháng 5/2019	
‣ Tổng số thuê bao điện thoại di động có phát sinh lưu lượng:	133,877,535
‣ Tổng số thuê bao điện thoại di động đang hoạt động chỉ sử dụng thoại, tin nhắn:	75,216,569
▪ Thuê bao trả trước:	70,448,710
▪ Thuê bao trả sau:	4,767,859
‣ Tổng số thuê bao điện thoại di động đang hoạt động có sử dụng dữ liệu:	58,660,966
▪ Thuê bao trả trước:	54,158,129
▪ Thuê bao trả sau:	4,502,837

Statistics are provided by Department of Telecommunication, Ministry of Information and Communication.
http://vnta.gov.vn/thongke/Trang/dulieuthongke.aspx

24

International Internet data volume 2019



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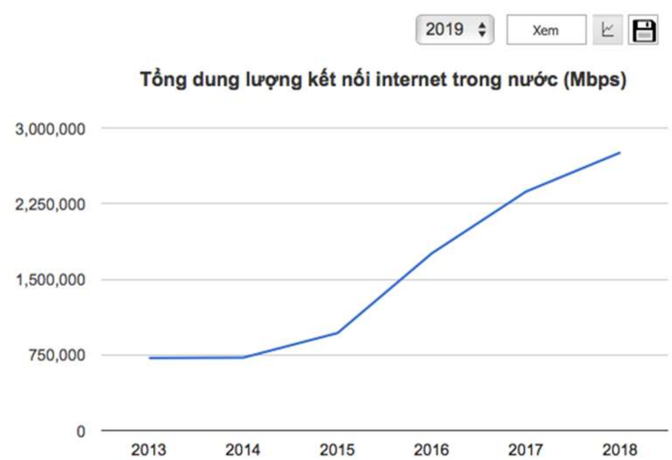
Domestic Internet data volume



Statistics are provided by Department of Telecommunication, Ministry of Information and Communication.
http://vnta.gov.vn/thongke/Trang/dulieuthongke.aspx

Internet management in Việt Nam

- VNNIC
  - Is responsible for managing the Internet domain name, address in Việt Nam;
  - Provides guidelines, statistics about Internet and participates in international activity about Internet.
- **VNIX: V**ietnam **N**ational **I**nternet e**X**change
  - switching system between national ISP.

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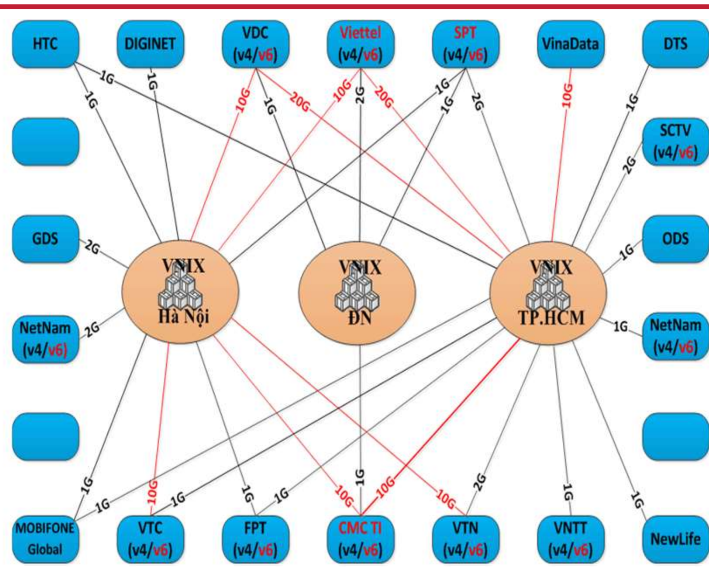
27

27



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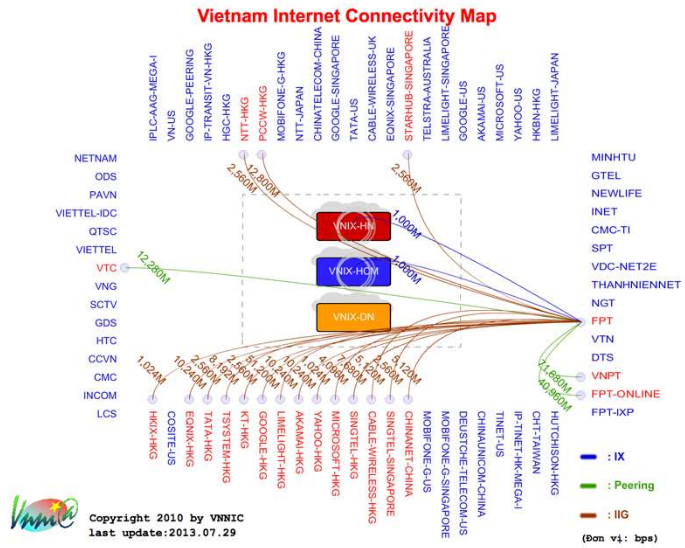
28

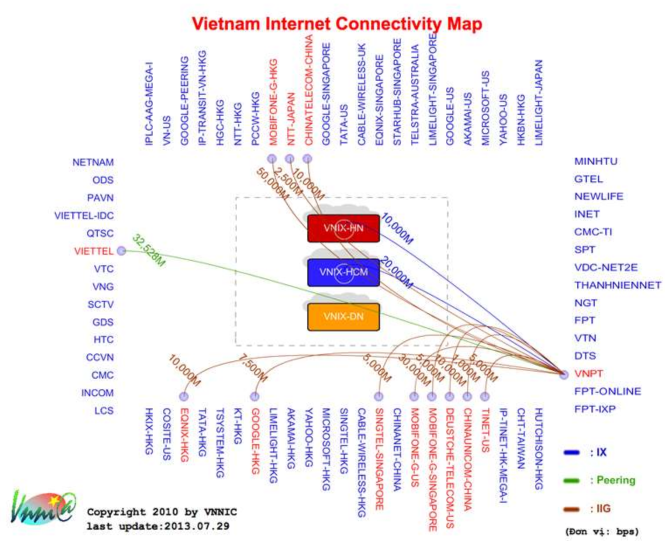
28

International connections



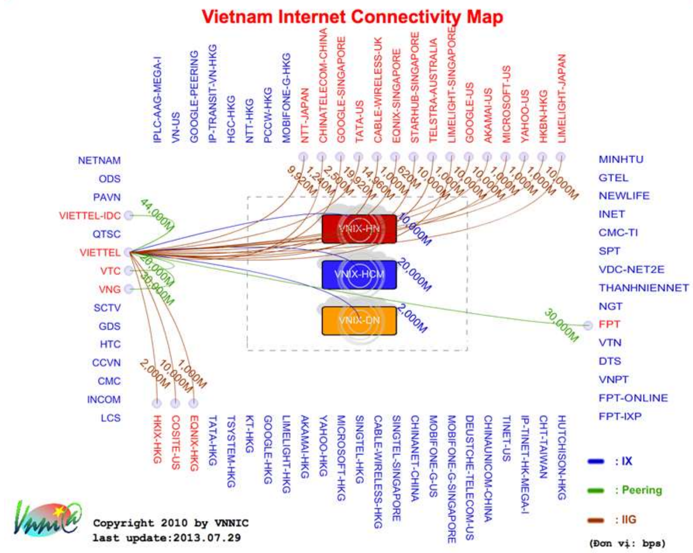
29

International connections



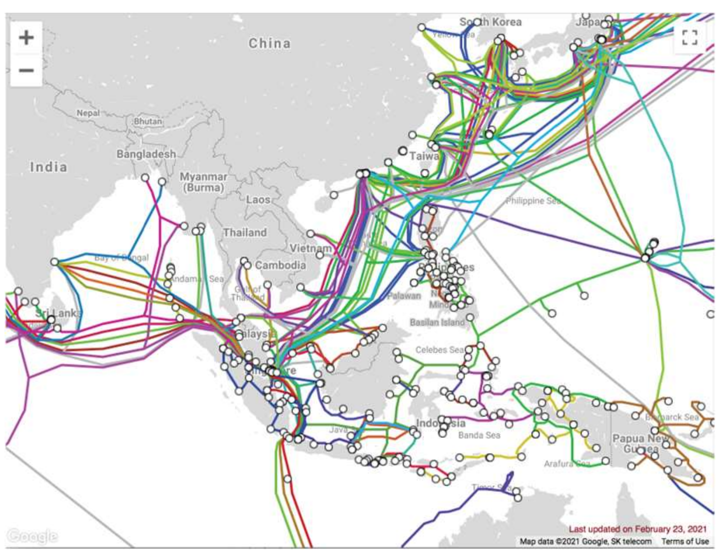
30

International connections



31

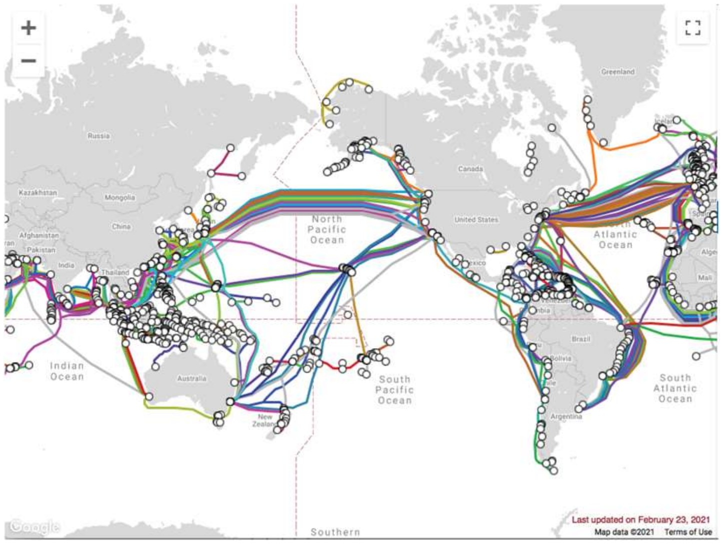
Optical fiber under the ocean



Source: https://www.submarinecablemap.com

32

Optical fiber under the ocean



Source: https://www.submarinecablemap.com

33

33

Concepts of computer networks

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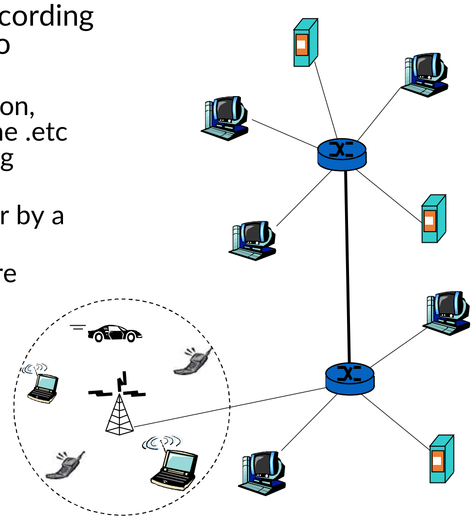
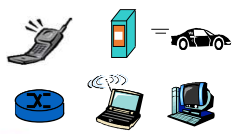
34

Concepts

- A set of computers/nodes connecting to each other according to an architecture in order to exchange data
  - Computer/node: workstation, server, router, mobile phone .etc with information processing capacity
  - They connect to each other by a media (wired or wireless)
  - Arcording to an architecture
- Different kind of computers

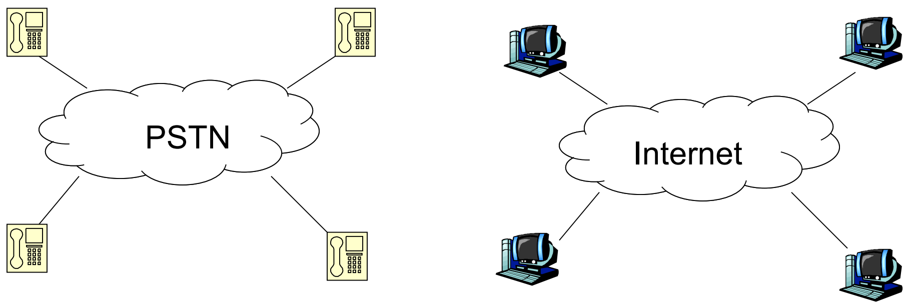




Example of networks

- Computer networks
  - The Internet
  - Ethernet
  - Wireless LANs: 802:11
  - ...
- Banking system (ATM networks)
- Traffic light networks
- Train networks (our new train systems in Hanoi and HCM city)
- Power, gas networks (in developed countries)

Centralized or distributed

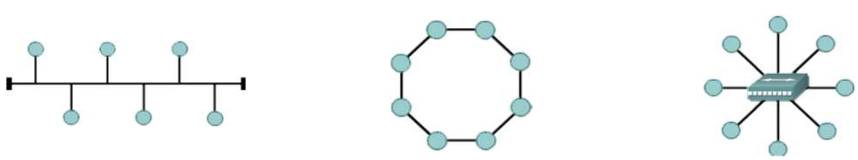


- Centralized: Network does everything

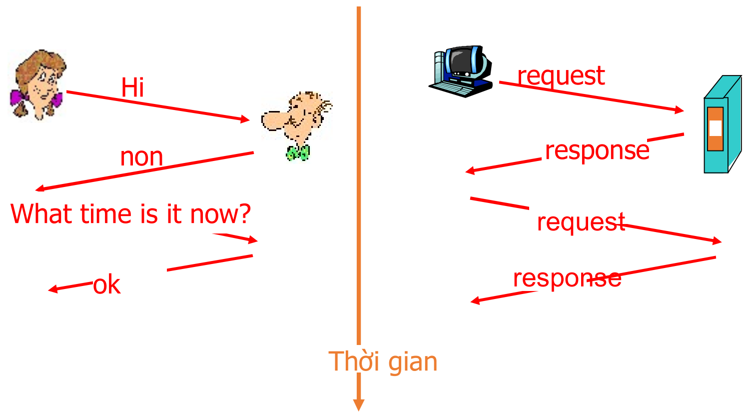
- Computer has stronger power
- Most functions are implemented at host

Network architecture

- Network architecture contain 2 aspects:
  - topology: the form that network nodes connects to each other
  - Protocol: language and procedure of communication between nodes.
- Topology
  - Bus, Ring, Star…



What is a protocol?



Protocol between human being: vocabulary, procedure

Protocol between machines

Network protocol

- A protocol defines communication rules between nodes
- Protocol defines:
  - Format of messages/ information to be exchanged between nodes.
  - Order of messages sending between entities/nodes
  - Action should be performed when an entity receives a message.
- Example of protocols running on the Internet: TCP, UDP, IP, HTTP, Telnet, SSH, Ethernet, …

Communication media

- Physical medium that can carry signal
- Classification:
  - Wired media: twisted pair, coaxial cable, optical fiber,…
  - Wireless media: radio wave, microwave, infrared wave,…
- Some characteristics:
  - Bandwidth (băng thông): width of the frequency band could be used for carrying signal
    - f_{min}: minimum frequency, f_{max}: maximum frequency
    - Bandwidth = f_{max} - f_{min}
  - BER – Bit Error Rate= nb of error bits/nb of transmitted bits)
  - Attenuation (suy hao): signal power decrement level

Computer network classification

- PAN – Personal Area Network
  - Scope: several metres
  - #users: few
  - To serve an individual
- LAN – Local Area Network):
  - Scope: few km
  - #users: few to hundreds of thousands
  - To serve a household, an organization

Computer network classification

- MAN – Metropolitian Area Network
  - Scope: hundreds of km
  - #users: Millions
  - To serve a metro, area
- WAN – Wide Area Network
  - Scope: thousands of km
  - #users: billions
- GAN – Global Area Network: over the world (ex: Internet)

43

LAN

- LAN (Local Area Networks):
  - Scope: a building, an office, an organization
  - Wireless LAN
    - VD: WIFI
  - Wired LAN
    - VD: Ethernet

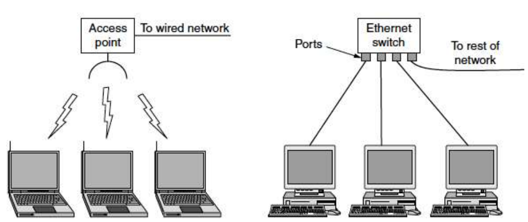


Figure 1-8. Wireless and wired LANs. (a) 802.11. (b) Switched Ethernet.

44

MAN

- Metropolitan Area Networks
  - Cover a city
  - Ex:
    - Television network
    - Backbone networks of ISP.

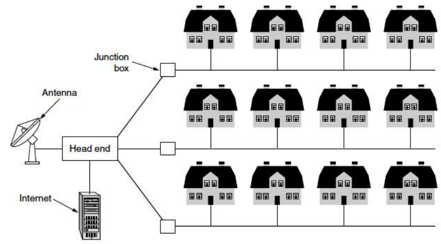


Figure 1-9. A metropolitan area network based on cable TV.

WAN

- Wide Area Networks
- Cover a large scope such as a country
  - Ex: network connecting different branches of a company
- Technology characteristics:
  - Using long distant lines to connect different parts of the network
    - Ex: Using public telephone network
    - Ex: using optical cable.

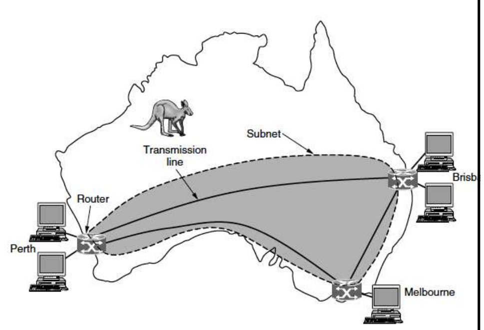


Figure 1-10. WAN that connects three branch offices in Australia.

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- Global Area Networks
- Interconnect different networks
- Cover many continentals.

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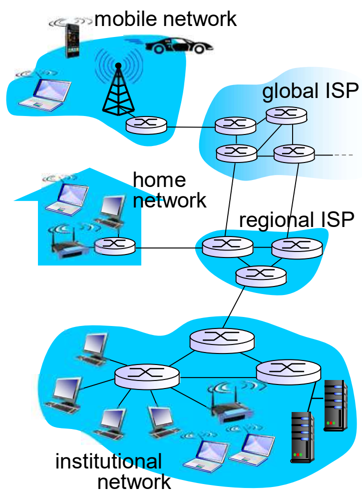
47

Internet

- Contain more than 5 billions devices
- 3.2 billion users (40%)
- Medium: optical fiber, twisted pair, Wimax, 3G…
- Transport ~3x109 GB data per day
- Services: Web, email, social networks, …



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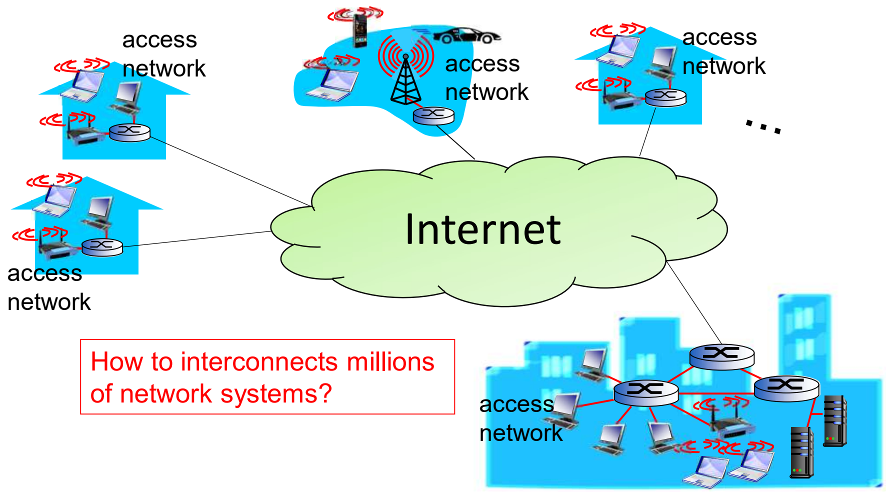
48

48

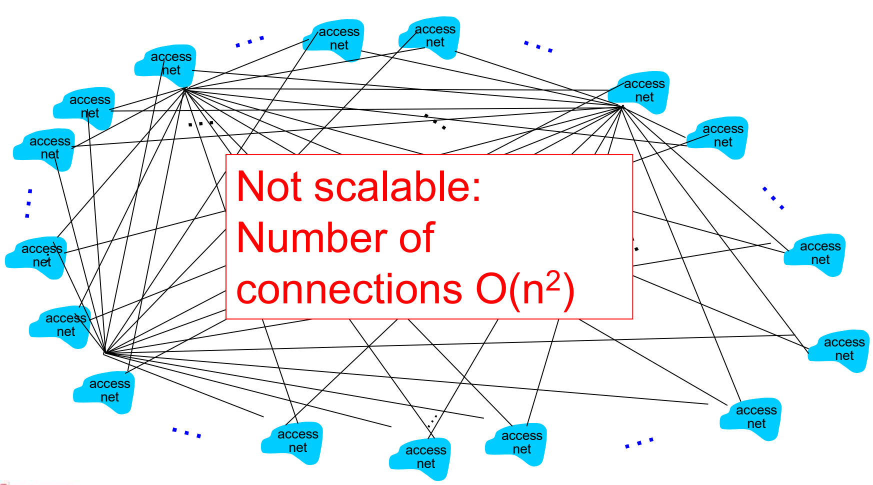
Internet

- Network of networks



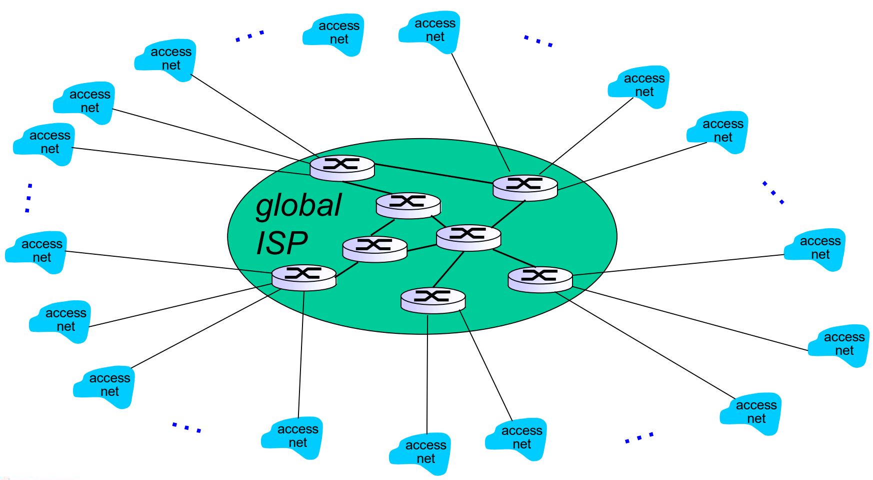
Internet: network of networks

- *Directly connect one network with all others?*



Internet: Network of networks(2)

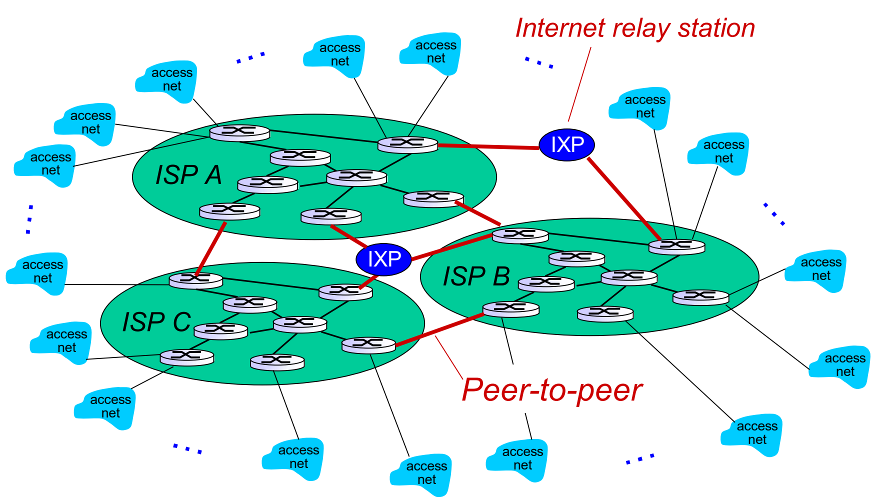
- *Connect each access network to a relay stations of a global ISP*



51

Internet: Network of networks(3)

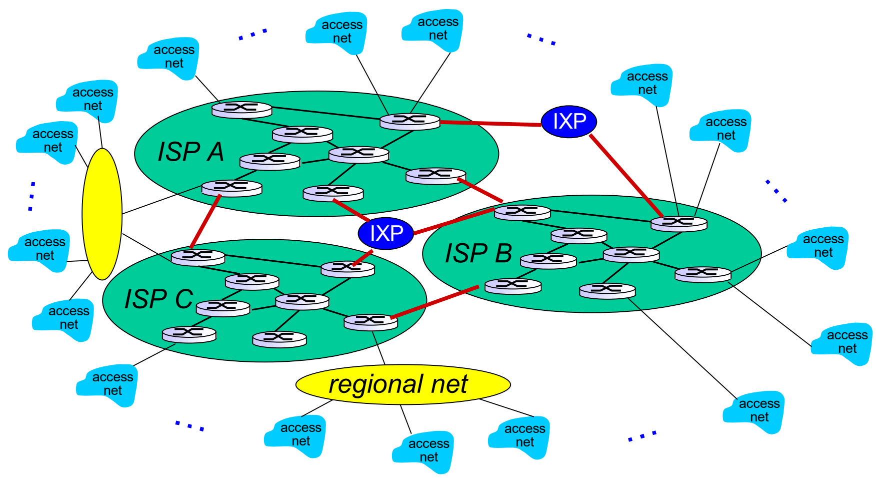
- *Add more ISP...*



52

Internet: Network of networks(4)

- *Add regional networks...*



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53

Transmission models

Packet switching vs. Circuit switching
Connection oriented vs. Connectionless

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54

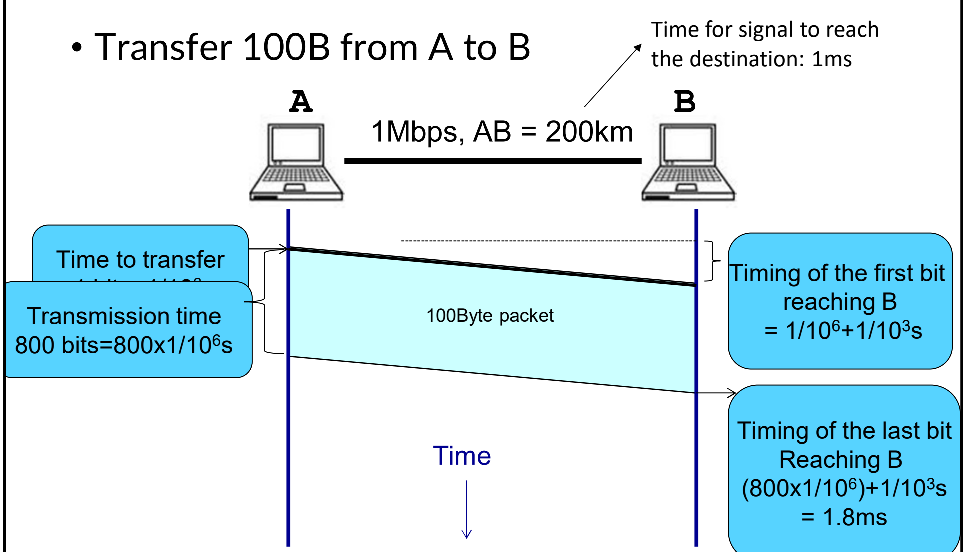
Problem

- Point-point connection between two hosts

- Connection parameters:
  - Bandwith - R: maximum amount of data transmitted within a time unit (bps – bit per second).
  - Example: optical cable has the bandwidth of 100Mbps.
  - Latency: transmission delay from A to B
    - Propagation delay: Connection length / speed of signal
    - Example: optical cable has the length of 10 km, speed of light (3×10^8m/s) ➔ $10 \times 10^3/(3 \times 10^8) \sim 3.333 \times 10^{-5} = 0.03333$ ms
    - Transmission delay: data size / bandwidth

Point-point connection

- Transfer 100B from A to B



Exercise

- Each packet with the size of 1000 bytes need to transfer through optical cable with the bandwidth of 100 Mbps. Cable length is 100km. Calculate
  - A) Time for the source to send a packet
  - B) Time for one bit to reach the destination (assume that the propagation speed is 200000 km/s)
  - C) Number of packages can be appeared on the transmission medium
  - D) The practical speed, if each sent bit need to be replied by an ACK bit by the destination

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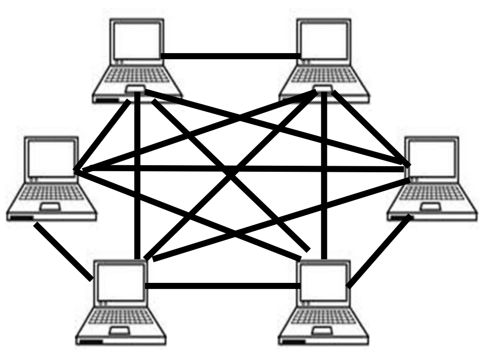
57

57

Connecting hosts
Direct links model

- Using direct links between all pairs of hosts
  - A link: a segment of medium without any processing unit in the middle
  - Weakness: too many links, distance limitation.



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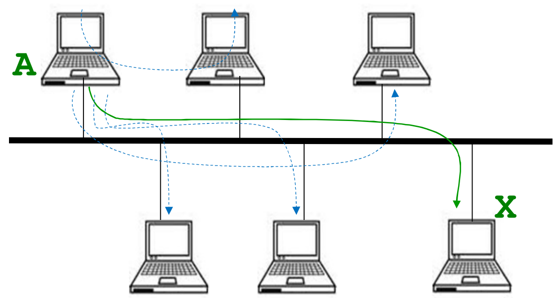
60

60

Connecting hosts
Bus model

- Point-to-multipoint:
  - Single communication medium is used for all hosts → broadcast communication
  - Weakness: long physical link, few hosts can communicate simultaneously

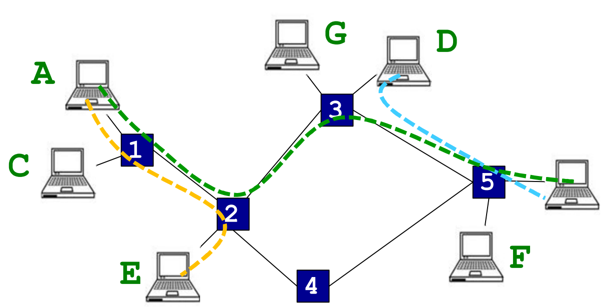


61

Connections between hosts

- Solution: switching network
  - Each host connects to be switching devices
  - Switching devices connect point-to-point and forward data to the destination
  - Share resources

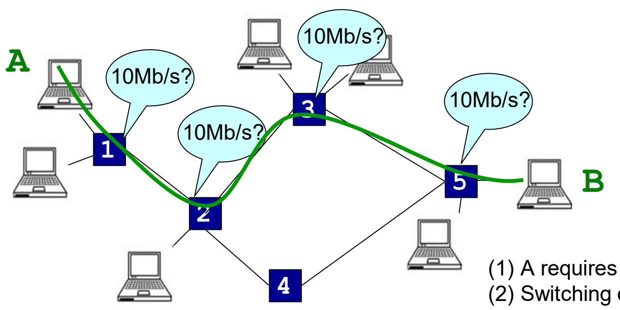
Question: How to determine the routes?
Answer: Routing



62

Circuit switching

- Circuit switching network: allocate resources for logical private channels between 2 network points

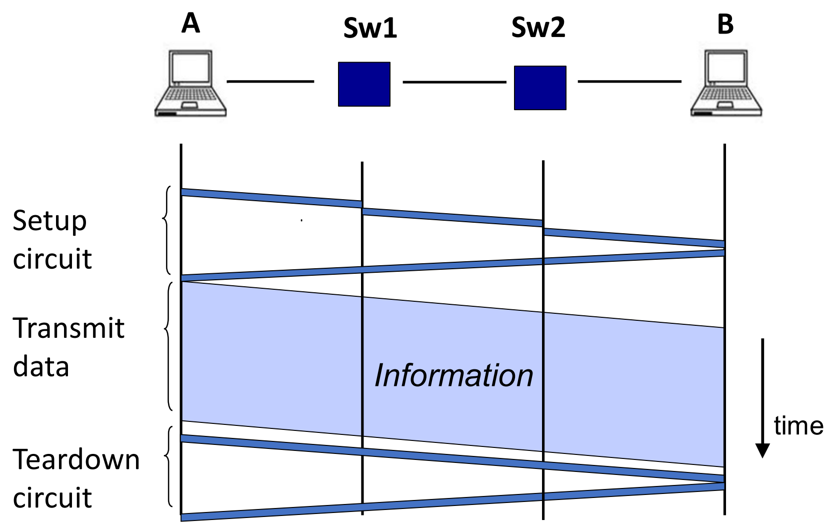


(1) A requires to establish a channel
(2) Switching devices establish the channel
(3) A starts data transmission
(4) A finish its transmission: send a signal to terminate the channel

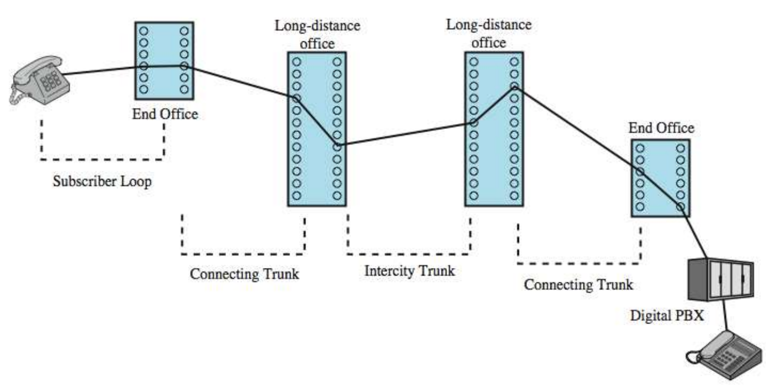
Circuit switching

- Resources (ex: bandwidth over a link) is dedicatedly assigned to a circuit. When the circuit is unused (no data is transmitted), no other circuits can use the resources.
- 3 phases of data transmission
  - Setup circuit: close ports of switches along the path
  - Transmit data
  - Tear down the circuit: release the closed switches
- Circuit switching guaranties that the circuits uses the whole available the bandwidth over each link for data transmission (good for audio/video transmission)
- Waste of bandwidth if the data transmission process does not consume the whole capacity of each link of the circuit.

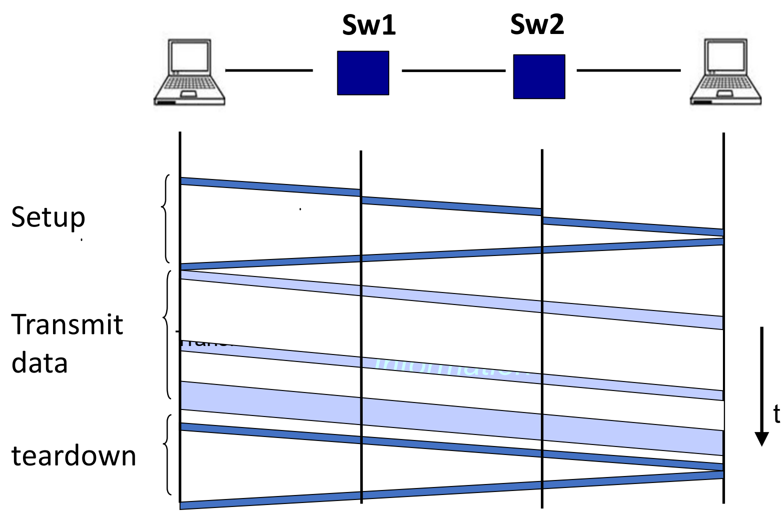
End-to-end data transmission time in circuit switching



Example of circuit switching : Public Switched Telephone Network PSTN

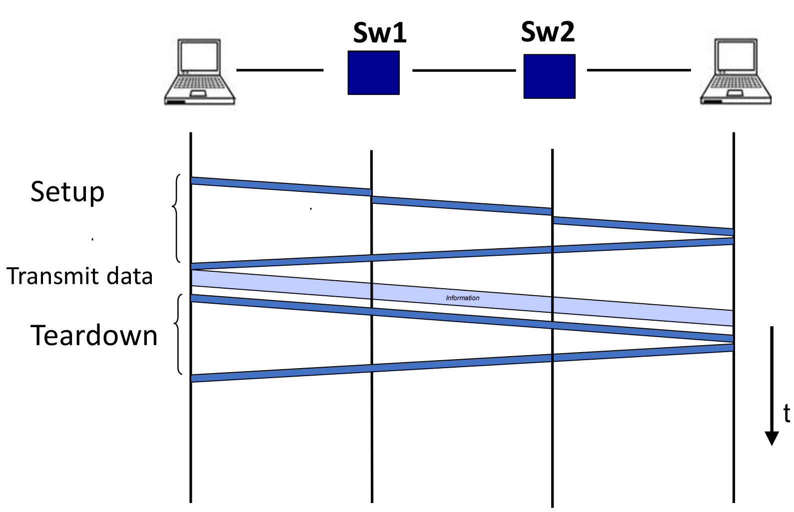


Cons : case of idle chanel ("blank" channel)



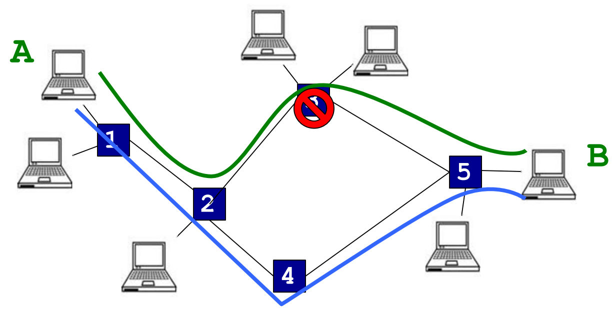
67

Cons : case of small channel



68

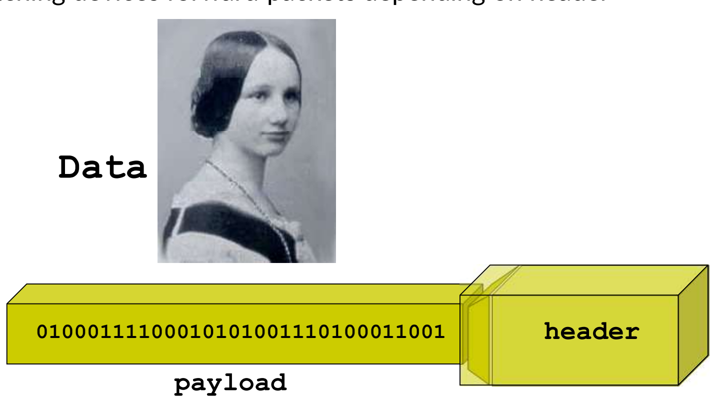
Cons: fault channel



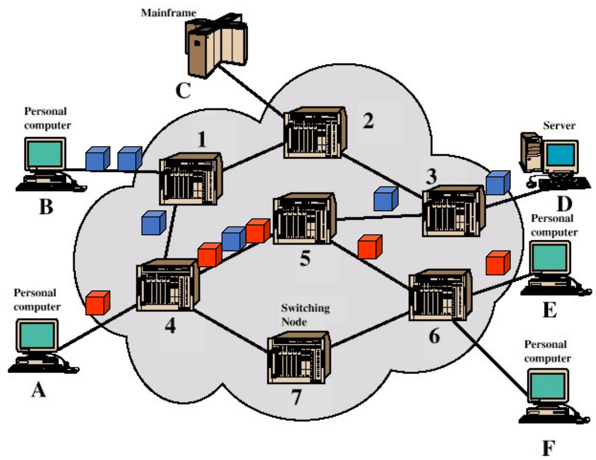
- Have to restart from the channel establishment stage if errors happen

Packet switching

- Data are splitted to packets
  - Header: address, order number
  - Payload data
- Switching devices forward packets depending on header



Example of packet switching



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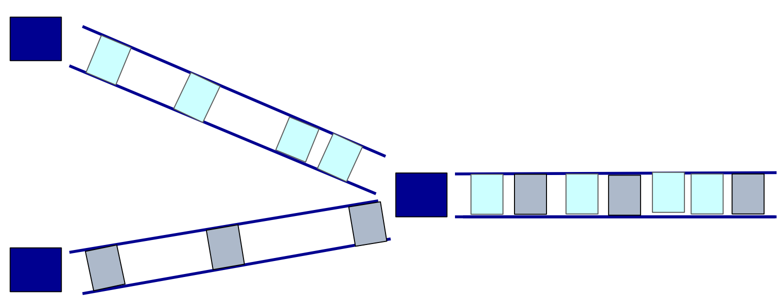
Packet switching

- Each packet can be processed independently
  - Each packet can have different paths to the destination, without the ordering
- Resources are shared for all connections
  - If resources are available, it can be used by any node



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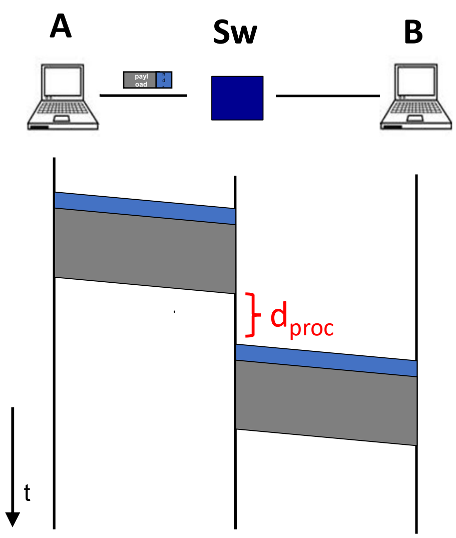
Transmission time in packet switching

- Switch forwards a packet only after receiving all the packet (store and forward)
- Switch needs time to process a packet (d_{proc}):
  - Checks error
  - Decides which ports to forward packet out
  - d_{proc} is usually smaller than transmission delay



Circuit vs packet switching

Example:

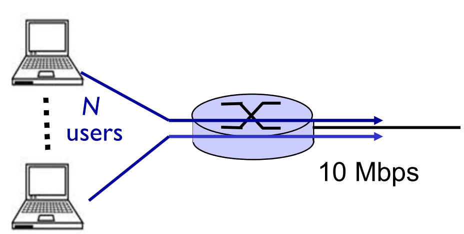
- Bandwidth 10 Mb/s
- Each connection of users:
  - Allocate 1 Mb/s
  - Time to use for data transmission: 10% of total time

❖ *Circuit switching:*

- Maximum 10 users can transmit data simultaneously

❖ *Packet switching :*

- Assume there are 30 shared users
- Probability of more than 10 users to send data at the same time? (~0.0003)



- *Binomial distribution:*

$P(x = k) = C_n^k p^k (1-p)^{n-k}$

- *What happen if there are more users?*

Switching performance

- Circuit switching network : Probability of all 10 users to send data at the same time:

$$P(k = 10) = C_{10}^{10} \times 0.1^{10} \times 0.9^{0} = 10^{-10}$$

- Packet switching network : Probability of more than 10 users to send data at the same time

$$P(k = 10) = C_{30}^{10} \times 0.1^{10} \times 0.9^{20} = 0.00037$$

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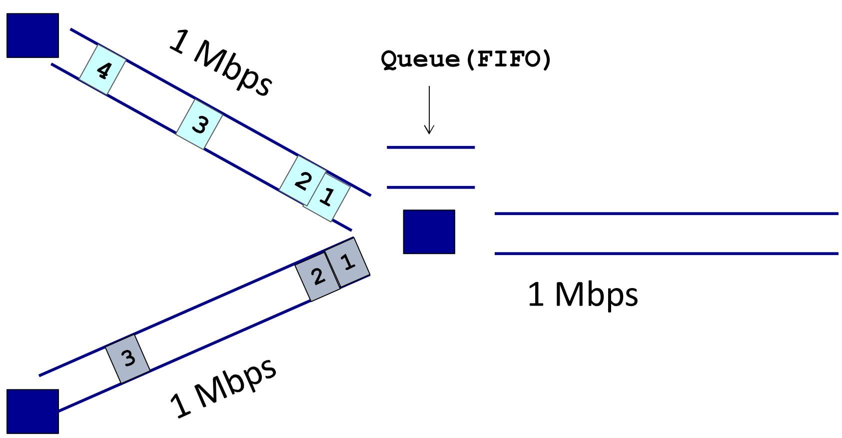
75

75

Queue



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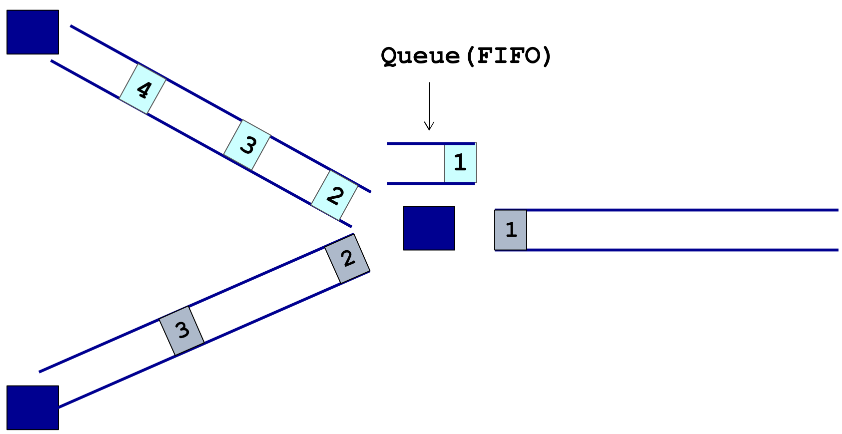
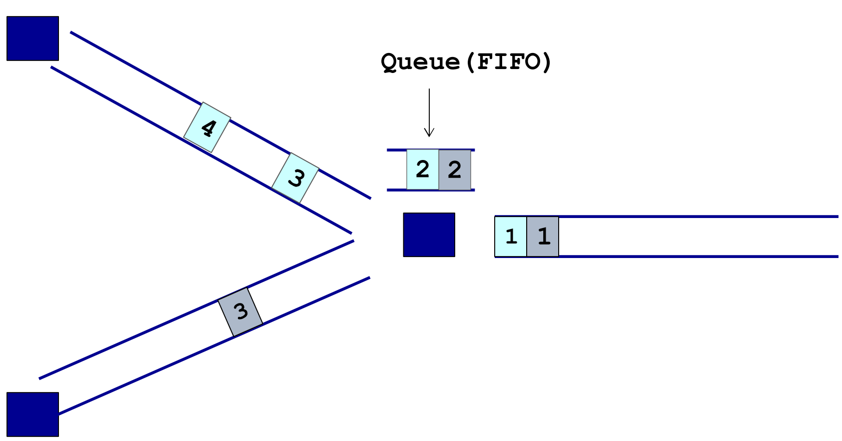
76

76

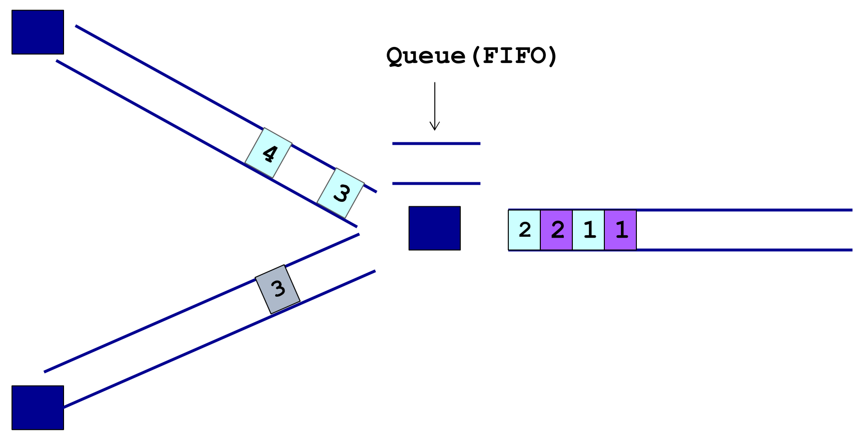
Queue



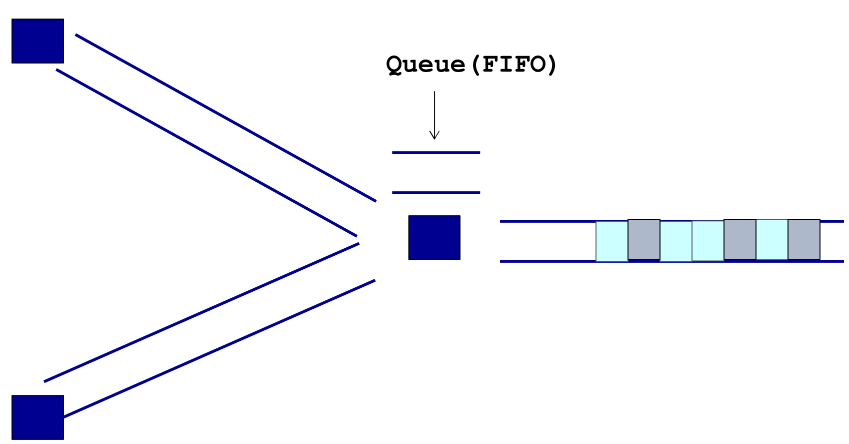
Queue



Queue

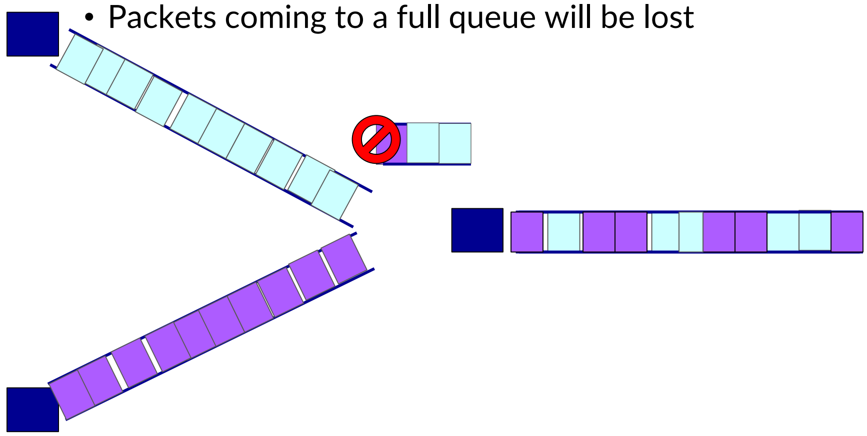


Queue



Packet loss

- Queue has its own limit
- Packets coming to a full queue will be lost



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3. Basic parameters in networks

- Bandwidth
- Throughput
- MTU (Maximum Transmission Unit)
- Latency
  - Delay on end nodes
  - Delay on intermediate nodes
  - Transmission delay
  - Propagation delay
- Package loss

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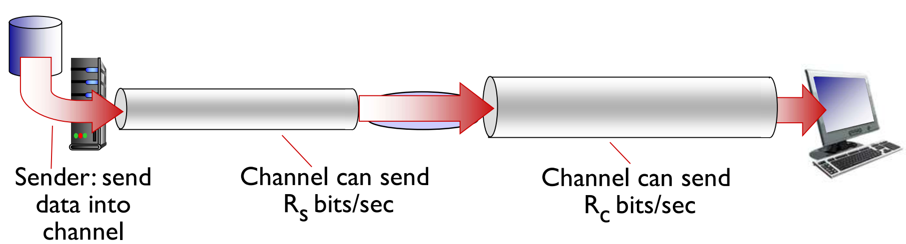
Bandwidth

- Bandwidth - R
  - In telecommunication: bandwidth= f_{max} - f_{min}
  - In computer networks: Maximum amount of data can be transmitted in a unit of time over a link (bps – bit per second).
  - Ex: optical fiber has bandwidth of 1000Mbps.

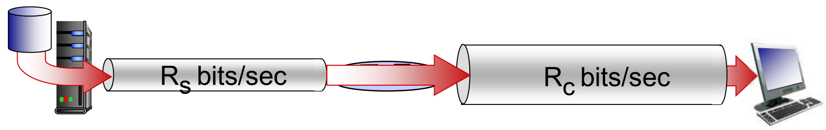
Thông lượng (throughput)

- *Throughput:* Speed (bits/sec) of data transmission at a time
  - *Instant:* throughput at a time
  - *Average:* average throughput during a period of time

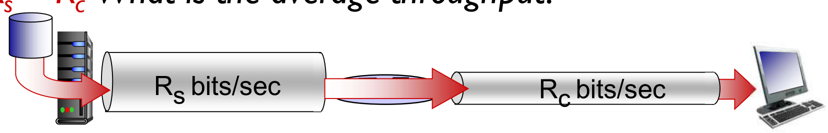


Throughput (cont.)

- $R_s < R_c$ *What is the average throughput?*



- $R_s > R_c$ *What is the average throughput?*

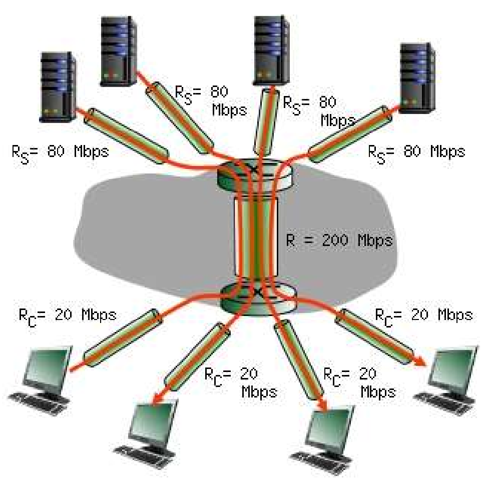


Nút thắt cổ chai (bottleneck)

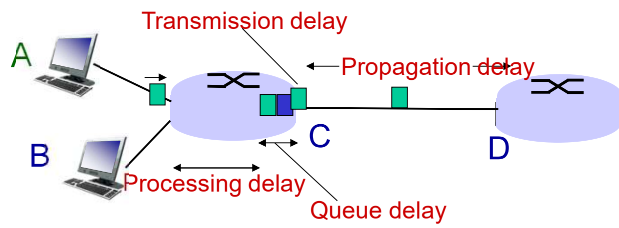
The point limits the bandwidth of the transmission connection

Bottleneck

- How to determine the bottlenecks?



Latency



$$\boxed{d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}}$$

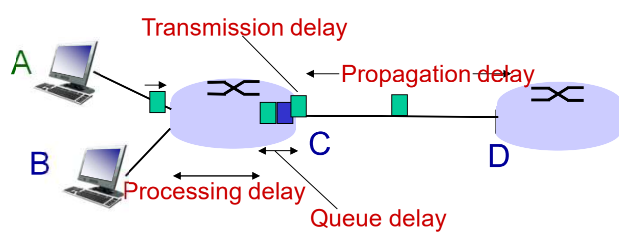
d_{trans}: transmission delay:
Time to send data out of a node

- L: data size(bits)
- R: bandwidth(bps)
- $d_{trans} = L/R$

d_{prop}: propagation delay
Time to propagate data from one end of link to the other

- d: length of link
- s: signal propagation speed in medium (ex: ~2×10^8 m/sec)
- $d_{prop} = d/s$

Latency



$$\boxed{d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}}$$

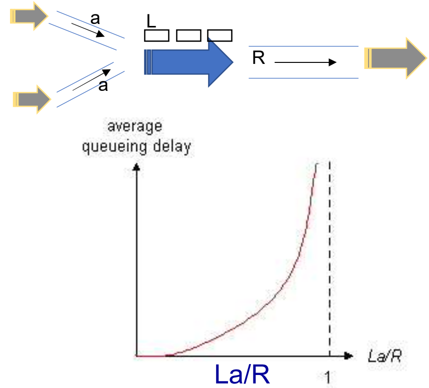
d_{proc}: processing delay

- Error check
- Identify out port
- Usually < μsec

d_{queue}: queue delay

- Time data stay in queue waiting for processing
- Depending on the amount of data in the queue.

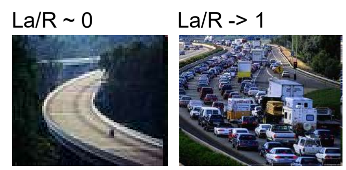
Queuing delay



- *R:* bandwidth(bps)
- *L:* size of packet (bits)
- a: coming speed of packet

- *La/R* ~ 0: small delay
- *La/R* → 1: high delay
- *La/R* > 1: extreme high delay (loss)

→ The problem of speed coordination on end-to-end connection



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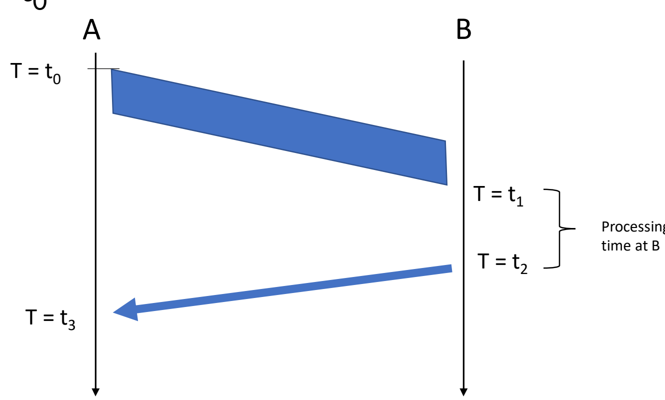
Round Trip Time (RTT)

- RTT: = $t_3 - t_0$



- One way delay: $t_1 - t_0$

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MTU

- Maximum Transmission Unit: maximum size of a packet can be sent on the connection
- Example: Ethernet has MTU of 1526 byte
- Why?
- Reason 1: reduce error rate
  - BER = error bits / total sent bits → constant
  - Ex: BER = 10^{-3} → send 1000 bits will yield 1 error bit
  - If L = 1000 bit → probability of a packet having error(s)?
  - If L = 100 bit → probability of a packet having error(s)?
- Reason 2: reduce the probability (or data size) of sending lost packets
  - Queue size: N byte
  - If L = 1000 byte: full queue → lost packet → resend it → send 1000 bytes
  - If L = 100 byte: full queue → ?
- Conclusion: reducing MTU will reduce the size of re-sending data

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But MTU cannot be too small

- Too small MTU will reduce performance of data transmission
- Explanation :
  - Packet: header + payload
  - Header: Constant
  - Performance:

$$H = \frac{payload}{header+payloa}$$

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Connection oriented vs connectionless communication

- Connection oriented communication:
  - Data is transmitted over a connection already setup
  - 3 phase of communication
    - Setup connection
    - Data transmission
    - Tear down connection
  - Connection setup allows to make sur that receiver is ready for the communication → more reliable
  - More control mechanism can be performed before the data transmission to enhance it QoS
- Connectionless
  - No connection is setup, there is only data transmission phase
  - Not reliable
  - "Best effort" QoS, sending data as quick as possible.

Summary

- Introduction to the course
- History of the Internet
- Concept of Computer Networks
- Architecture
  - Topology
  - Protocol
- Circuit switching vs. packet switching
  - Pros & cons

Lecture 2:
Basic concepts of computer networks

- Layer architecture
- OSI & TCP/IP reference model
- Addressing
- Domain name and conversion/resolution of domain name

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95

95

Layer architecture

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96

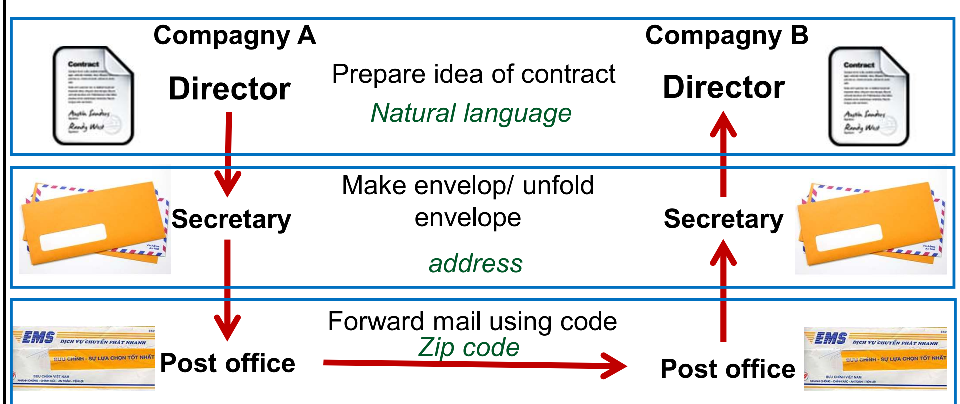
96

Devide and conquer principle

- Big work is divided into small tasks
- Assign some tasks to individuals
- Ex: Compagny A & B needs to discuss about a contract
  - Director of A,B: Identify the main points of the contracts & ask secretary to write down the contract.
  - Secretary:
    - Format the contract, put contract to envelope, write down the address of company B
    - Ask post office (VNPT) to send to company B
  - Post office:
    - Encapsulate the package/documents
    - Write down the received post office
    - Forward the envelop through several hub of post then to B

Example

- Parties at the same level performs similar tasks and use the same information communication methods.



Example of layers

Architecture with layers

Sound system

Player
Speaker
Amplifier

Architecture without layers

Cassette

All functionalities are put on the same box
When we want to upgrade: Upgrade the whole box

Advantage of layering systems

- For the complex system: principle of *" devide and conquer"*
- Allow to determine the responsibility of each layer and the relationship amongst them
- Allow to maintain and upgrade easily the system
  - Changes in some parts do not influence the other parts.
  - Ex: upgrade a media lecture from CD lecture to DVD lecture without the need to change speakers.

Reference models

OSI

TCP/IP

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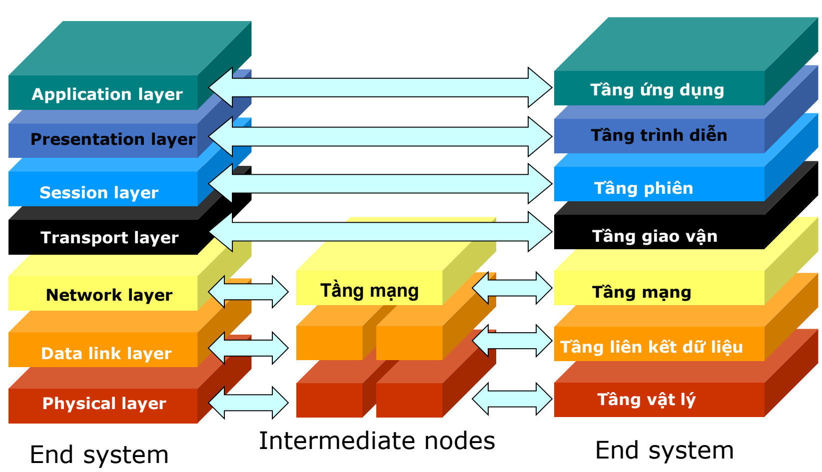
101

101

OSI - Open System Interconnection: 7 layers



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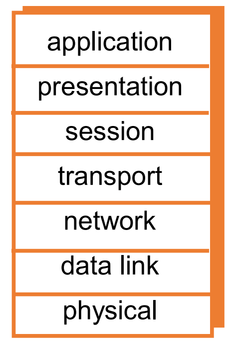
102

102

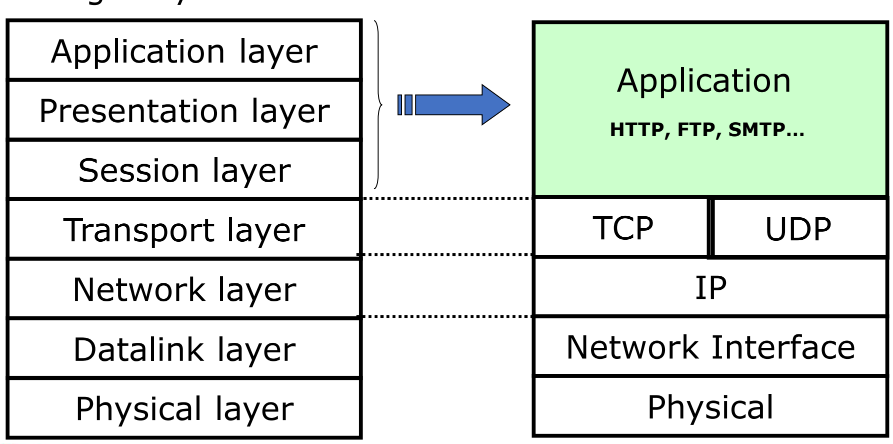
The main functionality of each layers

- Physical layer: Transfering bits "over medium"
- Datalink layer: Transfering data between direct connected elements in the networks.
- Network layer: Routing, forwarding data from the source to the distant destination
- Transport: Transmitting data between applications
- Session : synchronization, check-point, recovery of transmission process
- Presentation: data encoding, compression, data conversion…
- Application: Supporting communications between distant parts of an application.



Models OSI and TCP/IP

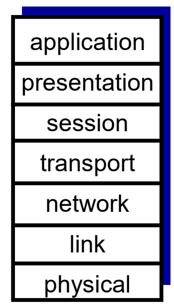
In the TCP/IP model of the Internet, the functionalities of 3 first layers are combined in a single layer.



ISO/OSI reference model

Two layers not found in Internet protocol stack!

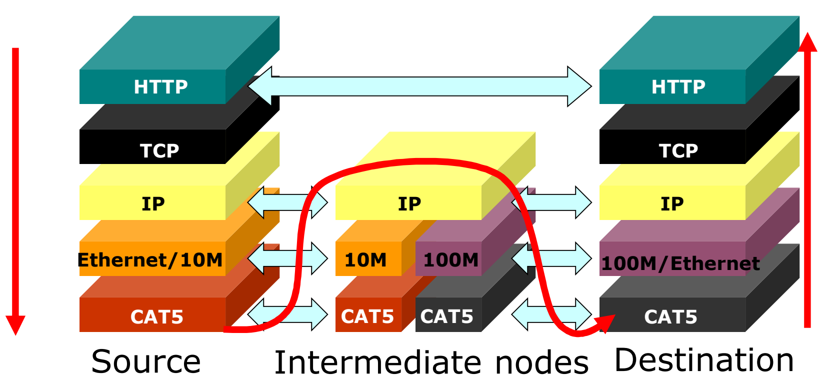
- *presentation:* allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- *session:* synchronization, checkpointing, recovery of data exchange
- Internet stack "missing" these layers!
  - these services, *if needed,* must be implemented in application
  - needed?



The seven layer OSI/ISO reference model

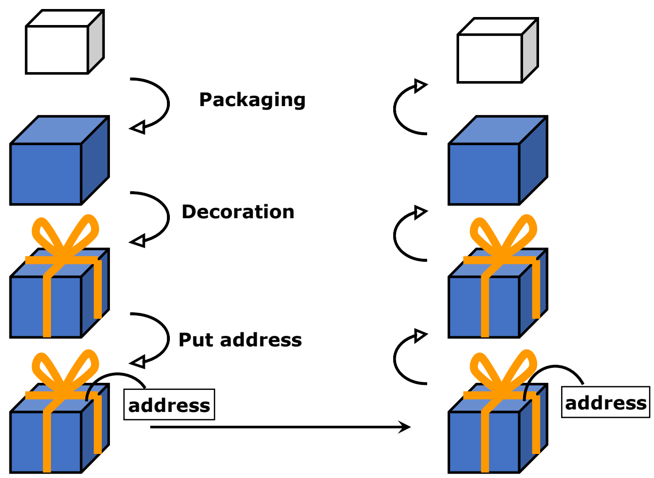
Layering model of the Internet

Example of data transmission from a source to a destination through intermediate nodes (router)



Data Encapsulation

Data encapsulation is similar to a packaging process for a gift.

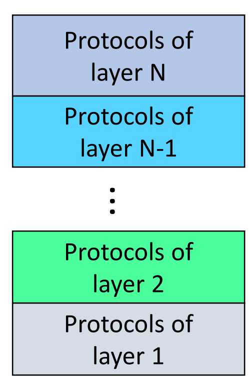


Protocol stack

- Functions are splitted to layers
- Each layer has multiple ways to implement its functions → multiple protocols

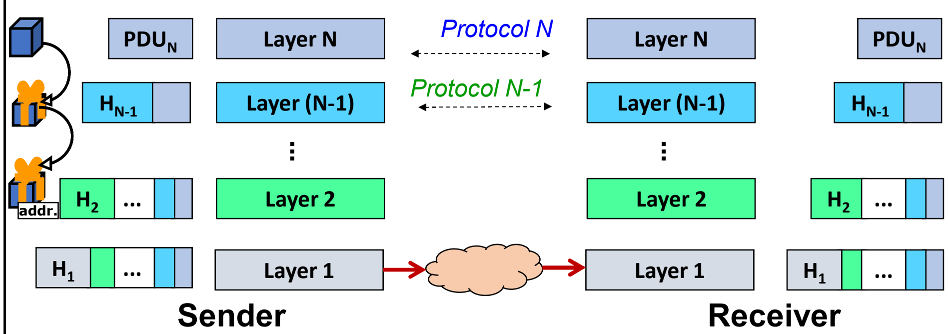
→Protocol stack: stack protocols based on layering systems

→Protocols at each layer include:
  - Call a service of the below layer
  - Provide service(s) for the upper layer



Data Encapsulation

- Sender side: Add header containing the information necessary for package processing at that layer, then send packet to the lower layer.
- Receiver side: Process data in the package according to information in the header, remove the header and send data to the upper layer.



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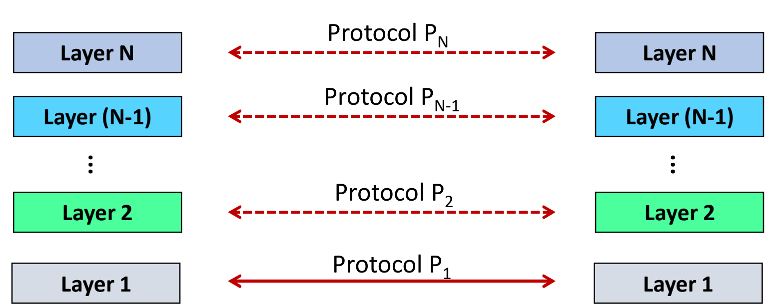
109

109

Data transmission in layering systems



- Layers (same level) on each side use the same protocol to exchange logical information
  - Two way to communicate logical information at the same logical level: connectionless or connected-oriented

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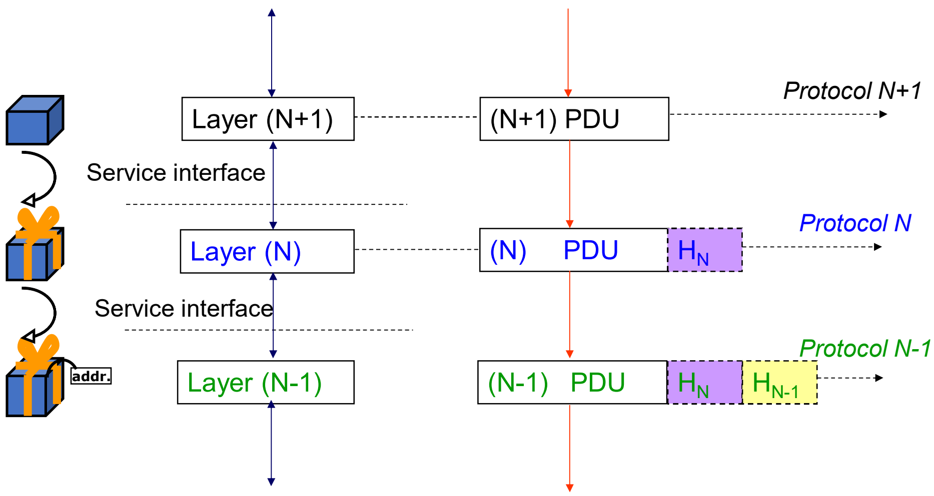
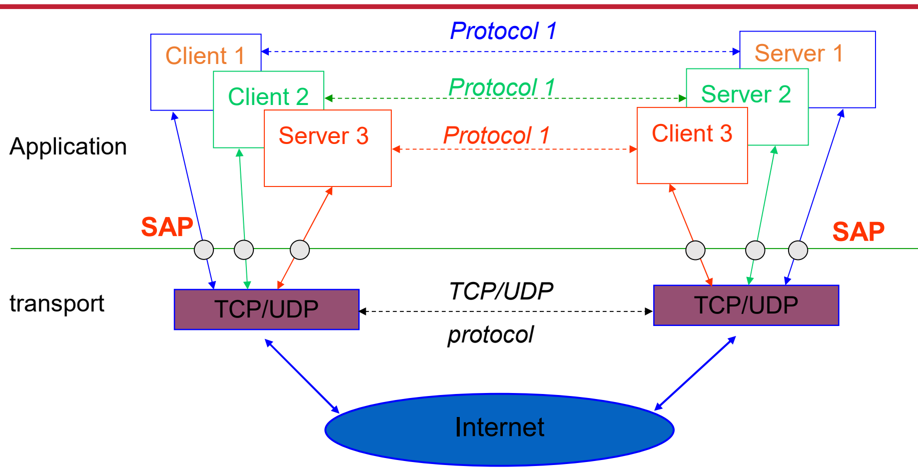
110

110

PDU: Protocol Data Unit



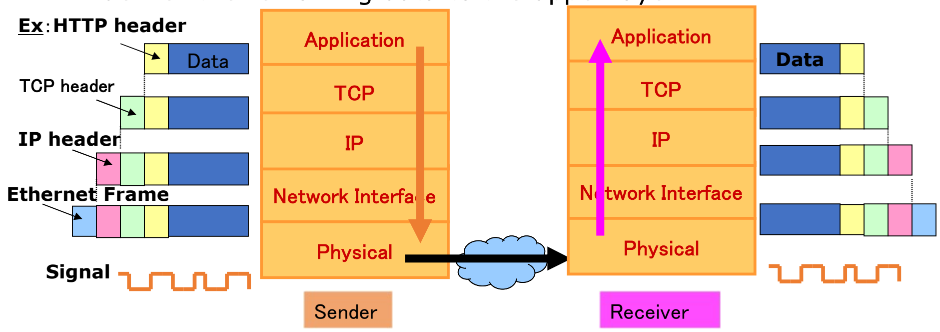
SAP: Service Access Point



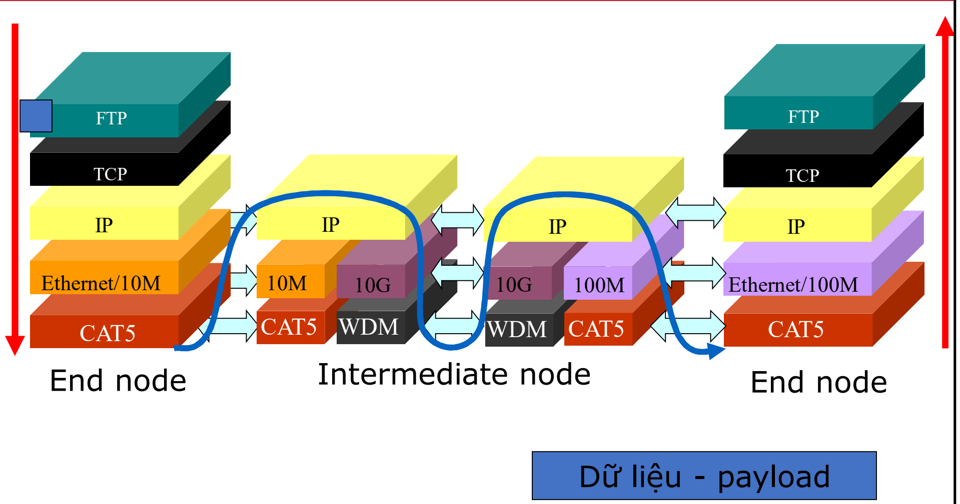
SAP: is a conceptual location at which one OSI layer can request the services of another OSI layer

The protocols TCP/IP and encapsulation process

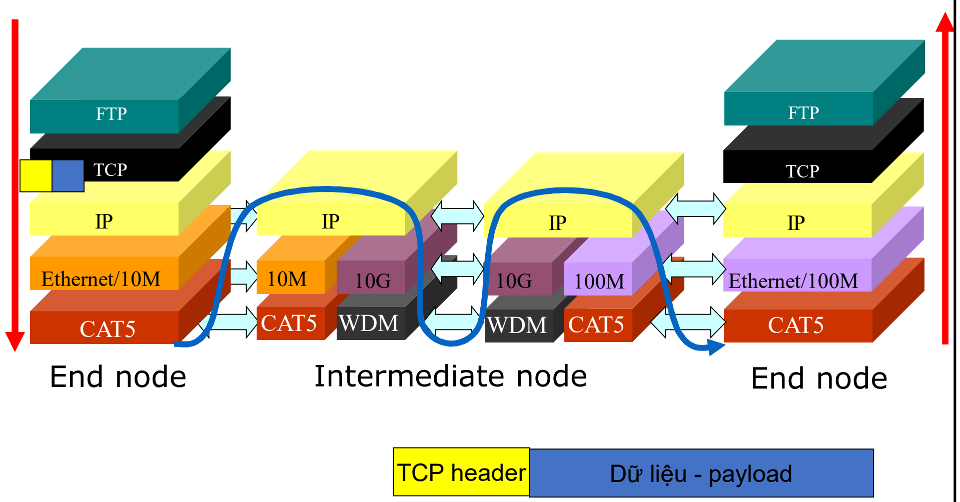
- At sender
  - Each layer add control information to the header of packet and transfer to the lower layer.
- At receiver
  - Each layer process packet according to the information of the header, then remove the corresponding header and deliver the remaining data to the upper layer.



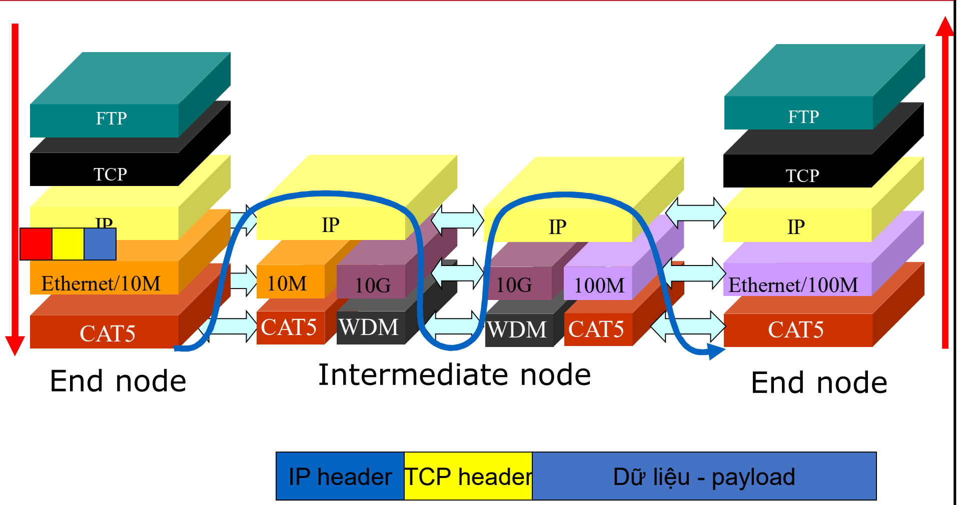
Protocol stack and encapsulation



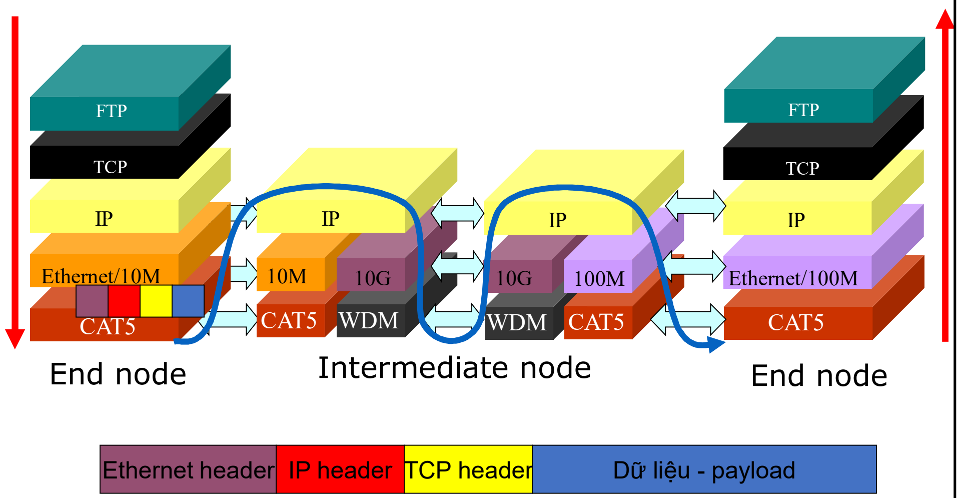
Protocol stack and encapsulation



Protocol stack and encapsulation



Protocol stack and encapsulation



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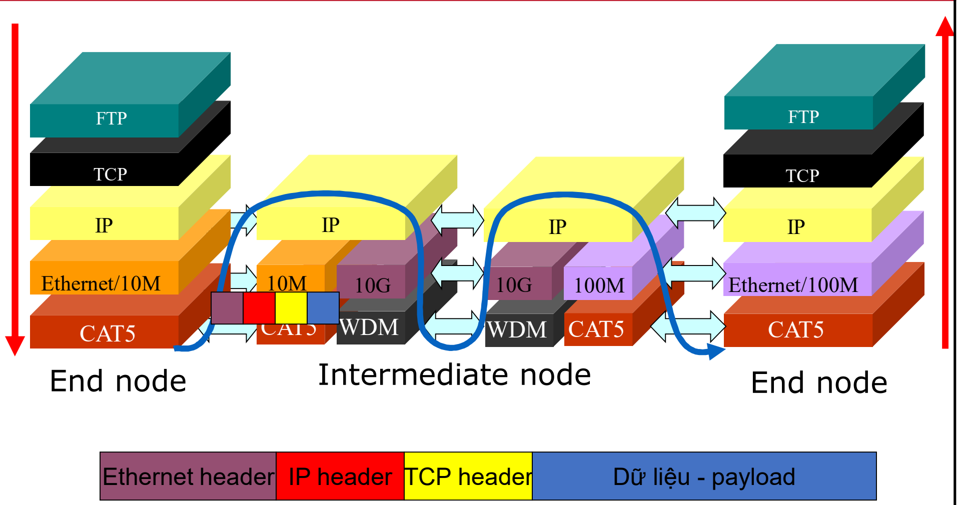
117

117

Protocol stack and encapsulation



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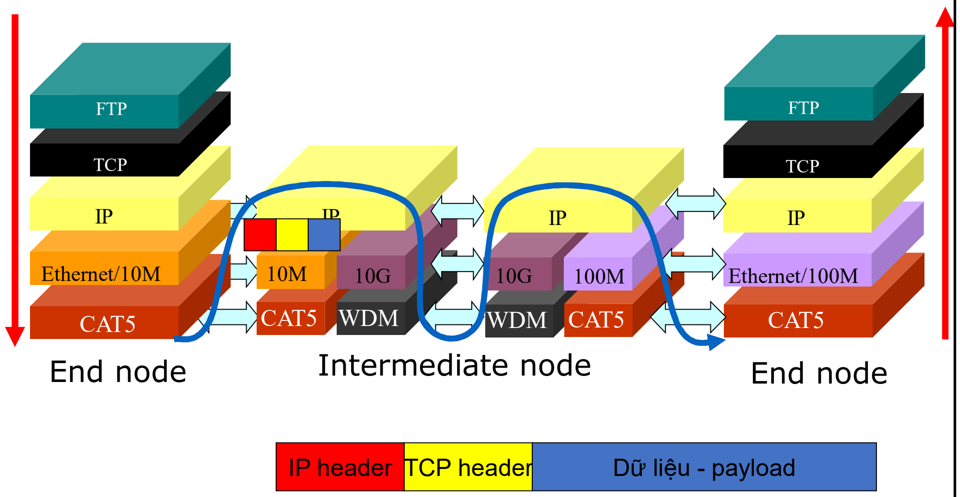
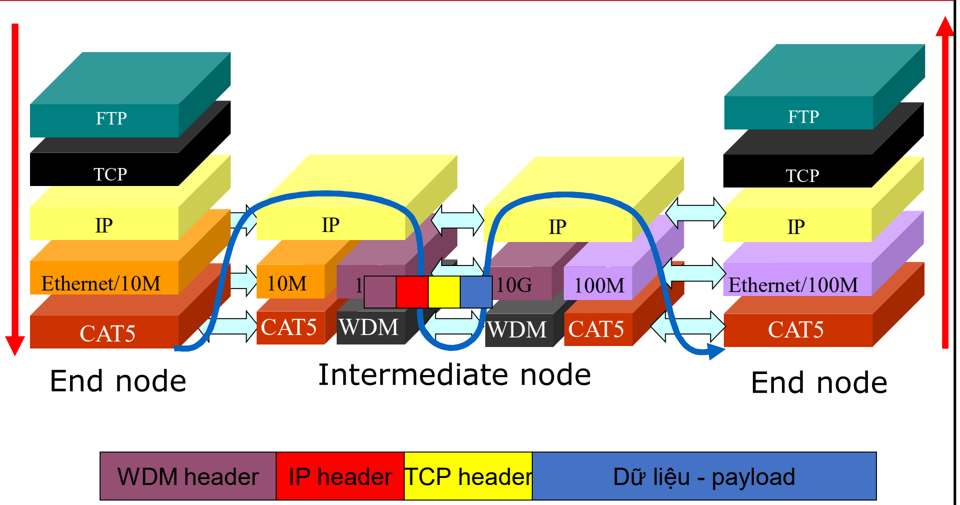
118

118

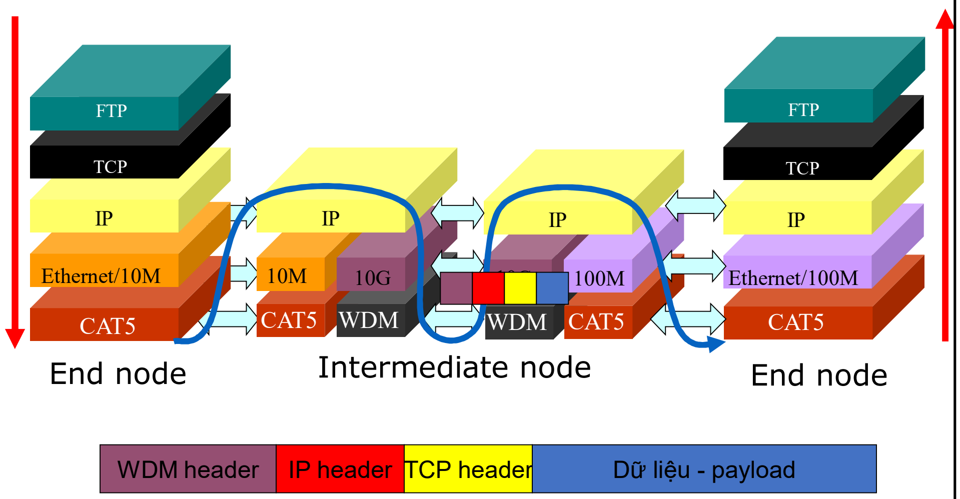
Protocol stack and encapsulation



Protocol stack and encapsulation

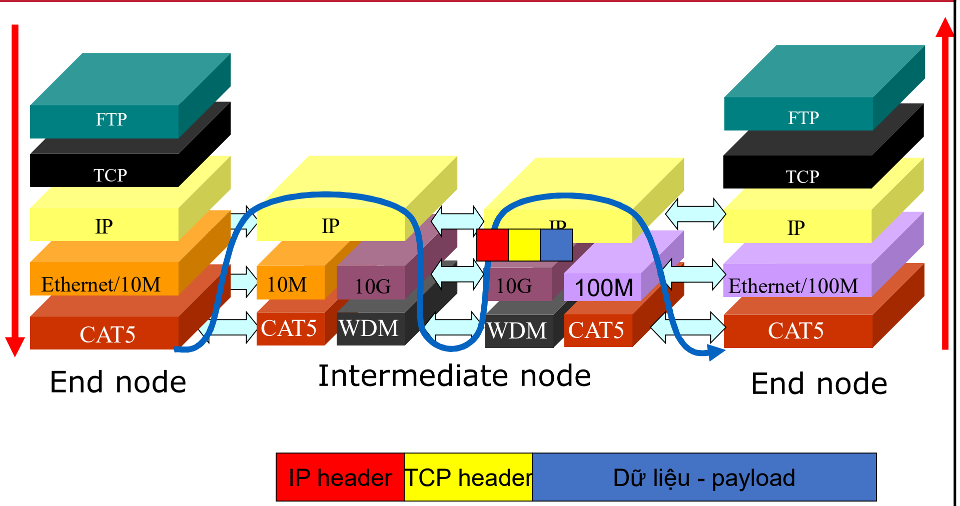


Protocol stack and encapsulation



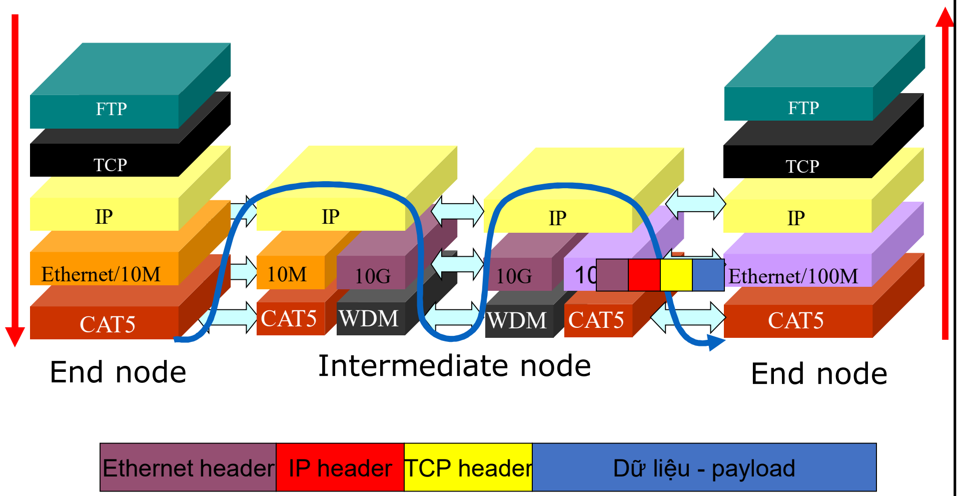
121

Protocol stack and encapsulation

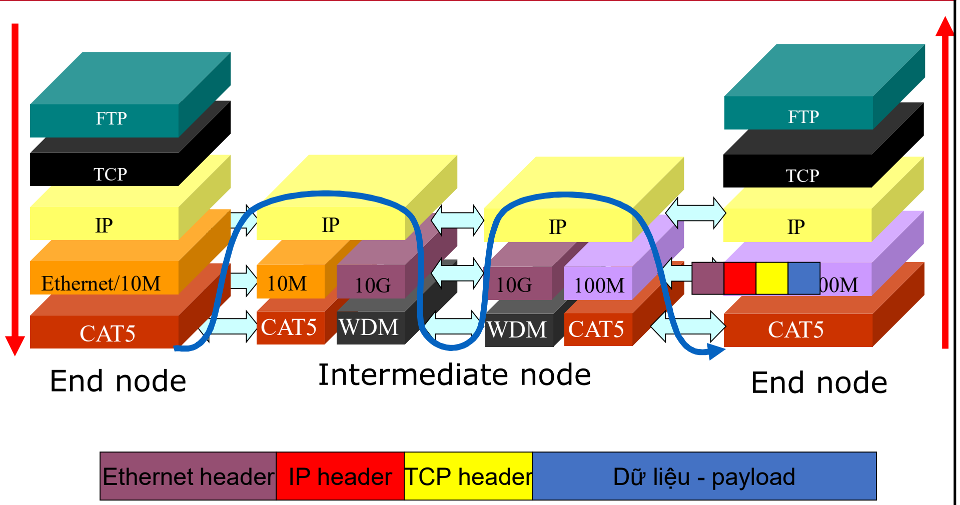


122

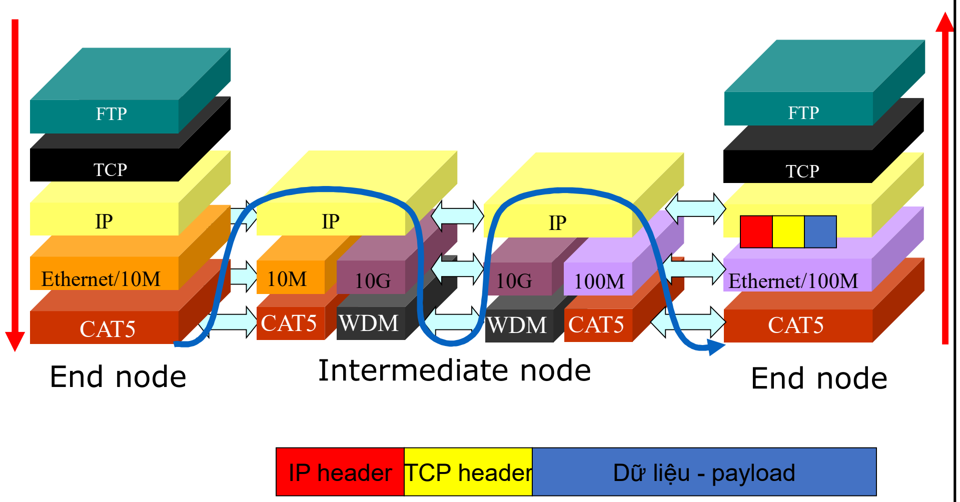
Protocol stack and encapsulation



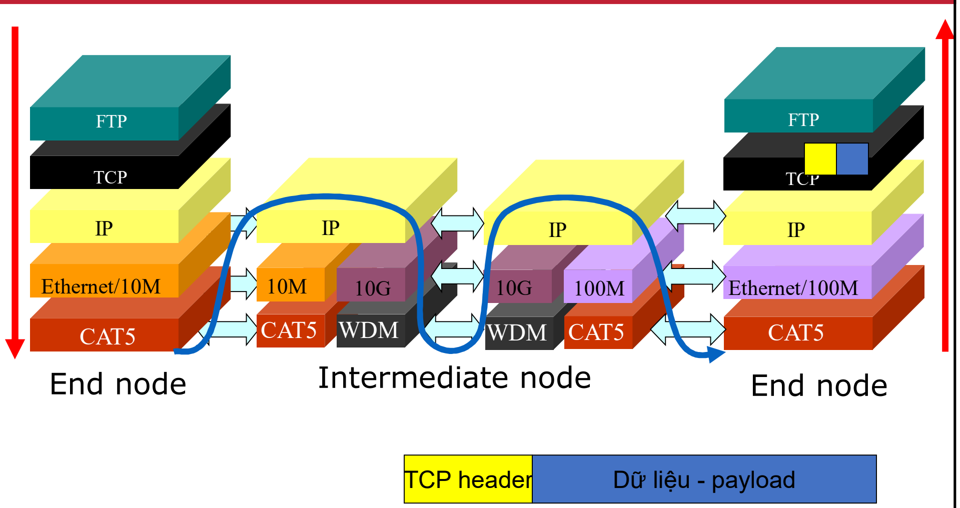
Protocol stack and encapsulation



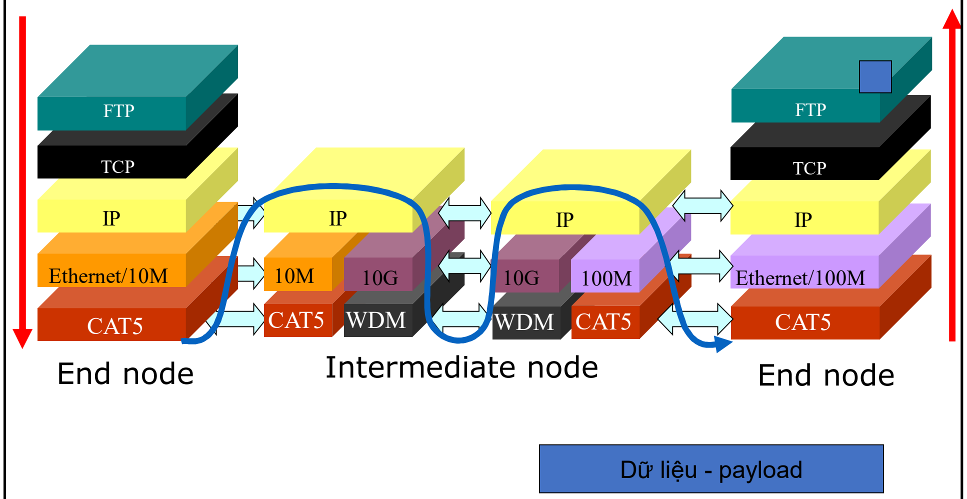
Protocol stack and encapsulation



Protocol stack and encapsulation



Protocol stack and encapsulation



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127

127

Summary: Advantage of layering architecture

- Layering architecture allows to divide the functionalities of networks into small components
- Layers are independent:
  - An upper layer makes use of the functionality of its right bellow layer but does not care about further layer.
- Extensibility/Scalability
- Flexible
  - It is possble to upgrade the communication system by upgrading the technology of some layers: Ex:
    - ADSL→FTTH
    - IPv4→IPv6
- Without layering:
  - Any change in the system requires changing the whole systems.

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128

128

Unicast, Multicast, Broadcast protocols

- Unicast protocol: control data to send to one destination node
- Multicast protocol: control data to send to multiple destination nodes
- Broadcast protocol: control data to send to all nodes

Identification in the Internet

MAC Address
IP Address
Port number

ONE LOVE. ONE FUTURE.

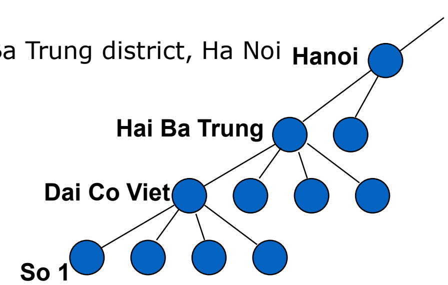
Identification

- Identification allows identify a person or an object
  - Name
    - Nguyen Thanh Ha
  - Address
    - 1 Dai Co Viet, Hai Ba Trung, Ha Noi
  - Telephone number
    - 8680896
  - Email
    - ha--xxx@it.hut.edu.vn

131

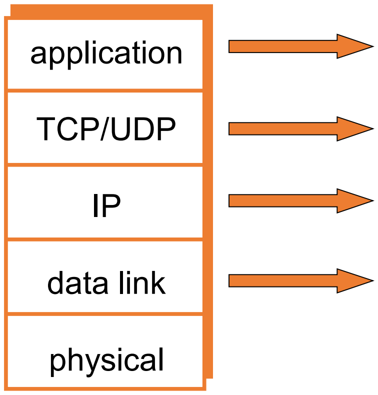
Identification

- Identification usually has hierarchical structure
  - Allow to manage efficiently a large addressing space
  - Scalability
- Example of hierarchy
  - Address
    - 1 Dai Co Viet street, Hai Ba Trung district, Ha Noi
  - Telephone number
    - +84-(4) 868-08-96



132

Identification in the Internet and the relationship between layers



Domain name, e.g., hust.edu.vn

Port Number, e.g. Port 80

IP address, e.g. 203.12.15.165

Physical address / MAC address
e.g. 00:11:24:79:8e:82

Addressing in the Datalink layer

- Physical address/ MAC address
  - Using in Datalink layer
  - Fixed on NIC (Network Interface Card)
  - Used for identifying machine in broadcast network segment.

HEX 00:11:24:79:8e:82

BIN 00000000 00010001 00100100 01111001 10001110 10000010

OUI (00000000 00010001 00100100) — Assigned by manufacture (01111001 10001110 10000010)

OUI (Organizationally Unique Identifier):
Each Manufacture have an some OUI unique

Addressing in the Internet

- IP address
  - Used in IP-Internet Protocol (network layer)
  - Value depends on the networks. Each network interface card should be assigned an IP address.
  - Used for identifying a machine in an IP network, example:
    - 133.113.215.10 (ipv4)
    - 2001:200:0:8803::53 (ipv6)

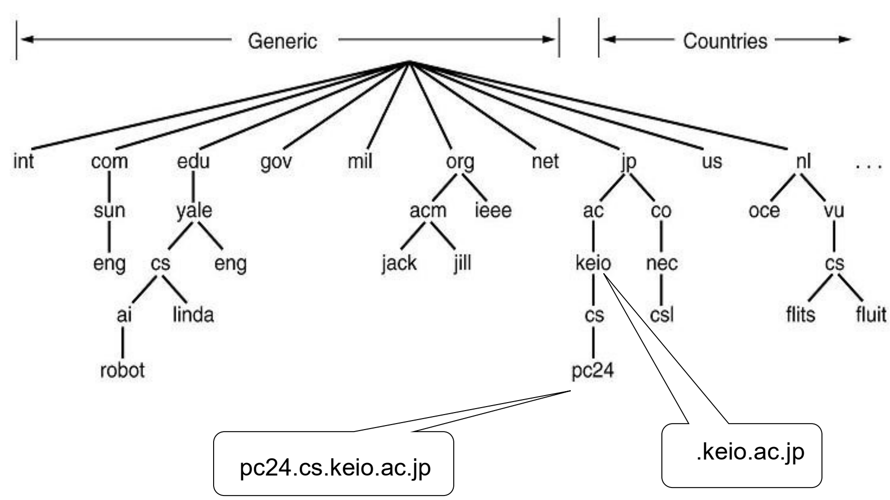
Addressing in transport layer

- Port number
  - On each machine, there may be several applications running.
  - Applications of the same machine are distinguished by port number.
  - An application instance in the internet is identified by the IP address of the host and port number on which it runs
  - Similar to the address of a room in a building
    - Buiding address: B1 Building, 1 Dai Co Viet, Ha Noi => similar to IP address
    - Room number 325 => Similar to port number
- E.g. HTTP runs on port 80, FTP runs on ports 20, 21 …
- http://bidv.vn:81

Addressing in Application layer

- Domain Name
  (FQDN: Fully Qualified Domain Name)
  - Domain name is the name given to a computer or a network using alphabet and numbers
    - www.keio.ac.jp
    - soict.hust.edu.vn

Domain name space



Domain name and IP address

- For sending data to a host/machine, the host must be identified
  - By an IP address
  - By a domain name (easy to be memorized by human)
- name
  - Variable length
  - easy to be memorized by human
  - Nothing to do with the location of the host
- IP address
  - Fixed length (32 bits or 128 bits)
  - Computer process address more easily
  - Used for routing purpose

203.162.7.194 ⇕ www.hust.edu.vn

www.hedspi.hut.edu.vn ⇕ 202.47.142.40

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139

139

Conversion/resolution of address

- Computer prefers numbers
- Human prefers names

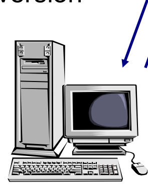
Need address conversion

User

I want to see the webpage www.hust.edu.vn

Domain name server

Need to access to 202.47.142.40



Webserver of www.hust.edu.vn 202.47.142.40

If user knows the IP address of the webserver, he can access via IP address from Browser

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140

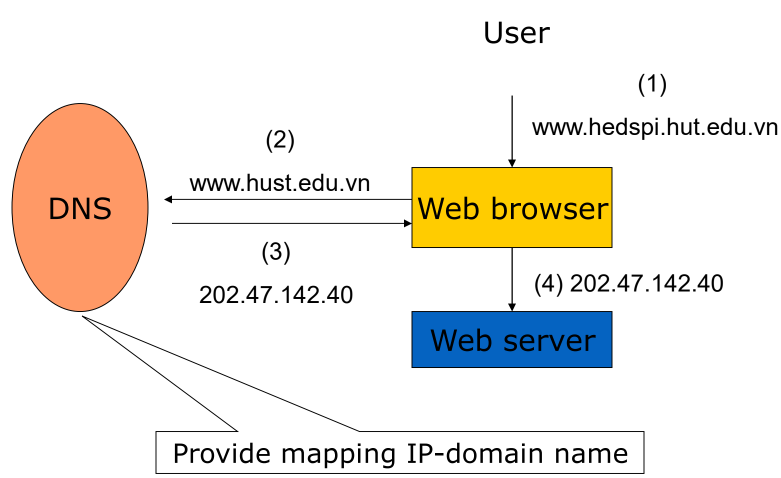
140

Address resolution/conversion

- Concept
  - Mechanism finding address IP from a domain name and vice versa.
  - There is no mathematical formula for this conversion.
- Domain name server (DNS)
  - Store the mapping of IP address and Domain name of the same host in a database
  - Answer requests to resolve IP addresses or domain names from users.
  - Widely used in the Internet

141

Example



142

Nslookup tools on Windows, Linux

- nslookup www.soict.hust.edu.vn
- Conversion “name⇔ IP address”

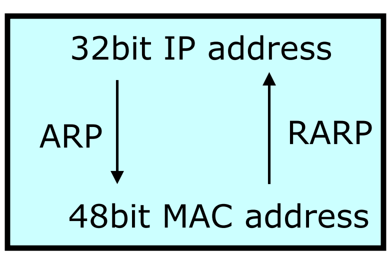
```
C:\>nslookup www.soict.hust.edu.vn
Server:
Address:  192.168.1.1

Non-authoritative answer:
Name:    www.soict.hust.edu.vn
Address:  202.191.56.68

C:\>
```


ARP Conversion of Mac and IP addresses

- Address Resolution Protocol
- MAC and IP are both used for identifying a NIC.
- ARP allows to find MAC address from IP address



Example: ARP table (on Windows)

```
C:\Documents and Settings\hongson>arp -a

Interface: 192.168.1.34 --- 0x2
  Internet Address      Physical Address      Type
  192.168.1.1           00-02-cf-75-a1-68     dynamic
  192.168.1.33          08-00-1F-B2-A1-A3     dynamic

C:\Documents and Settings\hongson>
```

IP address

MAC address

Summary

- Layer architecture
  - Why layering
  - Model TCP/IP vs. Model OSI
  - Encapsulation, PDU. SAP
- Addressing on Internet
  - Adress IP, MAC, domain name, port
  - Address conversion

Quizz

- What do the following objects identify
  - IP address
  - Transport port
  - Mac address
  - Domain name
- What identifies uniquely an application.
  - IP of the host running the application?
  - Transport port of the application?